

Foundations of General Relativity: From Flat Spacetime to the Schwarzschild Solution

A Mathematical and Geometric Approach to Gravitation

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Abstract

This thesis provides a comprehensive, mathematically rigorous review of General Relativity, tracing the theoretical development from the postulates of flat spacetime to the exact solutions of Einstein's field equations. We begin by formalizing Special Relativity using four-vector formalism and analyzing causality through the light cone structure. The mathematical framework of pseudo-Riemannian manifolds, tensor calculus, and differential forms is then systematically developed to describe curved spacetime. Utilizing the Einstein Equivalence Principle and minimal coupling, we derive the geodesic equation and Einstein's field equations, including discussions on the Newtonian limit, the Hilbert action, and the various energy conditions. Finally, we rigorously derive the Schwarzschild solution for a static, spherically symmetric vacuum. The causal structure of the Schwarzschild black hole is examined using Eddington-Finkelstein coordinates, and the classical experimental tests of General Relativity—specifically the precession of the perihelion and gravitational redshift—are analyzed. This work serves as a foundational synthesis of the geometric approach to gravitation, establishing the necessary groundwork for advanced studies in theoretical cosmology and high-energy physics.

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Chapter 1

Special Relativity and Flat Spacetime

1.1 Prelude

Core Concept of General Relativity (GR)

General relativity posits that gravity is not a traditional force, but a manifestation of spacetime curvature.

Newtonian Gravity: A Review

Force-Based Formulation:

$$F = -\frac{GMm}{r^2}\hat{e}_r \quad (1.1)$$

$$F = ma \quad (1.2)$$

Potential-Based Formulation: Using (1.1) and (1.2):

$$a = -\frac{GM}{r^2}\hat{e}_r \quad (1.3)$$

Gravitational flux for enclosed mass $M_{\text{enclosed}} = \int_v \rho dV$:

$$\text{Flux} = \oint_s a \cdot dA = -4\pi GM_{\text{enclosed}}$$

$$\oint_s a \cdot dA = -4\pi G \int_v \rho dV$$

By the divergence theorem and $a = -\nabla\Phi$, Poisson's equation relates potential Φ to mass density ρ :

$$\nabla^2\Phi = 4\pi G\rho \quad (1.4)$$

$$a = -\nabla\Phi \quad (1.5)$$

Generic Proof To Get Poisson's Equation

$$da = -G \frac{dm}{|r - r'|^2} \hat{e} * r - r' \quad (1.6)$$

$$dM = \rho(\mathbf{r}')dV' \quad (1.7)$$

$$\Phi = \oint_S \mathbf{a} \cdot d\mathbf{A} \quad (1.8)$$

Integrating acceleration over the source volume:

$$a(r) = \int *V_{\text{source}} - G \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^2} \hat{e} * (\mathbf{r} - \mathbf{r}')dV'$$

Evaluating the flux Φ given the surface integral is 4π when the source is inside:

$$\Phi = -4\pi G \int *V_{\text{source}} \rho(\mathbf{r}')dV' \quad (1.9)$$

Yielding:

$$\begin{aligned} \int_V (\nabla \cdot \mathbf{a})dV &= -4\pi G \int_{V_{\text{source}}} \rho(\mathbf{r}')dV' \\ \nabla \cdot \mathbf{a} &= -4\pi G \rho(\mathbf{r}') \\ \nabla^2\Phi &= 4\pi G \rho(\mathbf{r}') \end{aligned}$$

General Relativity: A Preview

Einstein's equation governs how energy-momentum curves spacetime:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu} \quad (1.10)$$

Free particles traverse geodesics:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0 \quad (1.11)$$

1.2 Space and Time, Separately and Together

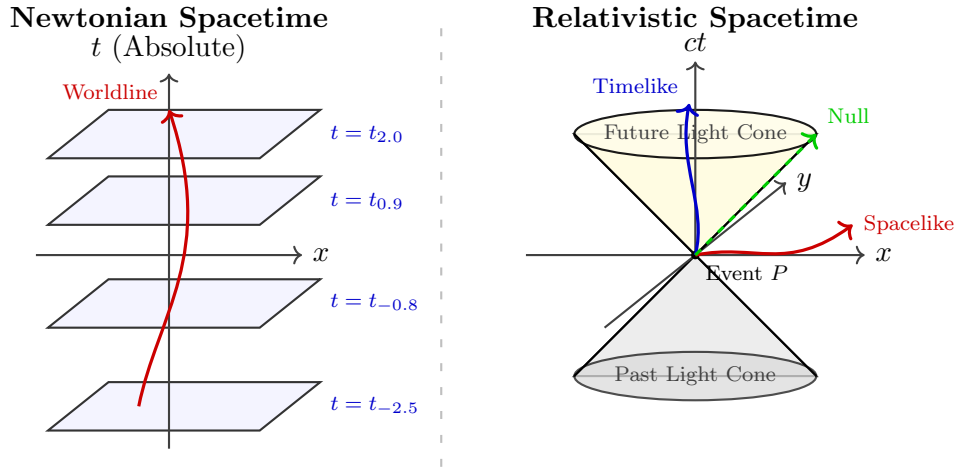
Special relativity replaces Newtonian absolute time with a unified four-dimensional spacetime.

Spacetime and Worldlines

Spacetime: 4D set of events. **Worldline:** 1D curve of a particle's trajectory.

Relativistic Spacetime: The Light Cone Structure

Special relativity eliminates absolute simultaneity. The invariant structure is the *light cone*, defining causality. No signal can travel faster than light.



The Invariant Spacetime Interval

The interval remains invariant for all inertial observers:

$$(\Delta s)^2 = -(c\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 = -(c\Delta t')^2 + (\Delta x')^2 + (\Delta y')^2 + (\Delta z')^2$$

Defining coordinates $x^\mu = (t, x, y, z)$ with $c = 1$, the flat spacetime (Minkowski) metric is:

$$\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1) \quad (1.12)$$

Causal Structure and Proper Time

- **Timelike:** $(\Delta s)^2 < 0$
- **Spacelike:** $(\Delta s)^2 > 0$
- **Null:** $(\Delta s)^2 = 0$

Proper time for timelike separations:

$$(\Delta\tau)^2 = -(\Delta s)^2 = -\eta_{\mu\nu}\Delta x^\mu\Delta x^\nu \quad (1.13)$$

General Trajectories

For any timelike trajectory $x^\mu(\lambda)$:

$$\Delta\tau = \int \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda$$

1.3 Lorentz Transformations

Translations and Linear Transformations

Translations:

$$x^{\mu'} = \delta^{\mu'} * \mu(x^\mu + a^\mu) \quad (1.14)$$

Linear Transformations (Rotations and Boosts):

$$x^{\mu'} = \Lambda^{\mu'} * \nu x^\nu \quad (1.15)$$

Condition for Lorentz Transformation

Invariance of $(\Delta s)^2$ requires:

$$\eta = \Lambda^T \eta \Lambda \quad \text{or} \quad \eta_{\rho\sigma} = \Lambda^{\mu'} * \rho \Lambda^{\nu'} * \sigma \eta_{\mu'\nu'} \quad (1.16)$$

This defines the **Lorentz group** $\mathbf{O}(3,1)$. The **Poincaré group** includes both Lorentz transformations and translations.

Explicit Transformation Matrices

Spatial Rotation (e.g., $x - y$ plane):

$$\Lambda^{\mu'}_{\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & \sin \theta & 0 \\ 0 & -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (1.17)$$

Boost (along x -direction, rapidity ϕ):

$$\Lambda^{\mu'}_{\nu} = \begin{pmatrix} \cosh \phi & -\sinh \phi & 0 & 0 \\ -\sinh \phi & \cosh \phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (1.18)$$

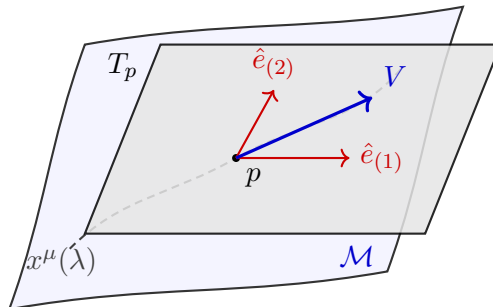
Setting $v = \tanh \phi$ and $\gamma = 1/\sqrt{1-v^2}$, this becomes:

$$t' = \gamma(t - vx) \quad (1.19)$$

$$x' = \gamma(x - vt) \quad (1.20)$$

1.4 Vectors

Vectors reside in the **tangent space** T_p at a given point p .



Basis and Transformation

A vector $A = A^\mu \hat{e}_{(\mu)}$. The tangent vector to $x^\mu(\lambda)$ is $V^\mu = dx^\mu/d\lambda$. Vector components transform via Λ :

$$V^{\mu'} = \Lambda^{\mu'} * \nu V^\nu \quad (1.21)$$

Basis vectors transform inversely to keep the geometric vector invariant:

$$\hat{e} * (\nu') = \Lambda^\mu * \nu' \hat{e} * (\mu) \quad (1.22)$$

1.5 Dual Vectors (One-Forms)

The **cotangent space** T_p^* contains dual vectors ω , linear maps from $T_p \rightarrow \mathbb{R}$.

Basis and Action

Basis dual vectors satisfy $\hat{\theta}^{(\nu)}(\hat{e}_{(\mu)}) = \delta_\mu^\nu$. Expanding $\omega = \omega_\mu \hat{\theta}^{(\mu)}$:

$$\omega(V) = \omega_\mu V^\mu \in \mathbb{R} \quad (1.23)$$

Transformation

Dual vector components transform via the inverse matrix:

$$\omega_{\mu'} = \Lambda^\nu * \mu' \omega * \nu \quad (1.24)$$

Gradient

The gradient $d\phi = \partial_\mu \phi \hat{\theta}^{(\mu)}$ is a natural dual vector:

$$\frac{\partial \phi}{\partial x^{\mu'}} = \Lambda^\nu * \mu' \frac{\partial \phi}{\partial x^\nu} \quad (1.25)$$

$$\partial * \mu \phi \frac{dx^\mu}{d\lambda} = \frac{d\phi}{d\lambda} \quad (1.26)$$

1.6 Tensors

A (k, l) tensor multilinearly maps k dual vectors and l vectors to $\mathbb{R}$.

Tensor Components

Constructed via tensor products $\otimes$, components transform as:

$$T^{\mu'_1 \dots \mu'_k * \nu'_1 \dots \nu'_l} * \nu'_1 \dots \nu'_l * l = \Lambda^{\mu'_1 * 1} * \mu_1 \dots \Lambda^{\nu_l * l} * \nu'_l * l T^{\mu_1 \dots \mu_k * \nu_1 \dots \nu_l} \quad (1.27)$$

Spacetime Tensors

- **Metric** $\eta_{\mu\nu}$: $\eta(V, W) = \eta_{\mu\nu} V^\mu W^\nu$.
- **Inverse Metric** $\eta^{\mu\nu}$: $\eta^{\mu\rho} \eta_{\rho\nu} = \delta_\nu^\mu$.
- **Levi-Civita** $\tilde{\epsilon}_{\mu\nu\rho\sigma}$: ± 1 for even/odd permutations.
- **EM Field Tensor** $F_{\mu\nu}$:

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix} = -F_{\nu\mu} \quad (1.28)$$

1.7 Manipulating Tensors

Contraction: Summing one upper and one lower index ($S^{\mu\rho}{}_{\sigma} = T^{\mu\nu\rho}{}_{\sigma\nu}$). **Raising/Lowering:**

$$T^{\alpha\beta\mu} * \delta = \eta^{\mu\gamma} T^{\alpha\beta}{}_{\gamma} * \gamma\delta \quad (1.29)$$

$$V_{\mu} = \eta_{\mu\nu} V^{\nu} \quad (1.30)$$

Trace: Contraction of indices $X = X^{\lambda}{}_{\lambda} = \eta^{\mu\nu} Y_{\mu\nu}$.

1.8 Maxwell's Equations

Inhomogeneous Equations: Combine $\partial_i F^{0i} = J^0$ and $\partial_j F^{ji} - \partial_0 F^{0i} = J^i$:

$$\partial_{\mu} F^{\nu\mu} = J^{\nu} \quad (1.31)$$

Homogeneous Equations:

$$\partial_{[\mu} F_{\nu\lambda]} = 0 \quad \implies \quad \partial_{\mu} F_{\nu\lambda} + \partial_{\nu} F_{\lambda\mu} + \partial_{\lambda} F_{\mu\nu} = 0 \quad (1.32)$$

Manifest covariance ensures frame-independent validity.

1.9 Energy and Momentum

Four-Velocity and Four-Momentum

$$U^{\mu} = \frac{dx^{\mu}}{d\tau}, \quad \eta_{\mu\nu} U^{\mu} U^{\nu} = -1 \quad (1.33)$$

$$p^{\mu} = mU^{\mu}, \quad p_{\mu} p^{\mu} = -m^2 \implies E = \sqrt{m^2 + p^2} \quad (1.34)$$

Four-Force

$$f^{\mu} = \frac{dp^{\mu}}{d\tau} = qF^{\mu}{}_{\lambda} U^{\lambda} \quad (1.35)$$

Energy-Momentum Tensor $T^{\mu\nu}$

$T^{\mu\nu}$ measures four-momentum flux. **Dust:** $N^{\mu} = nU^{\mu}$ and $T^{\mu\nu}_{\text{dust}} = \rho U^{\mu} U^{\nu}$. Under arbitrary boost:

$$N^{\mu'} = n\gamma(1, -v_x, -v_y, -v_z)^T \quad (1.36)$$

$$T^{\mu'\nu'} = \rho\gamma^2 \begin{pmatrix} 1 & -v_x & -v_y & -v_z \\ -v_x & v_x^2 & v_x v_y & v_x v_z \\ -v_y & v_y v_x & v_y^2 & v_y v_z \\ -v_z & v_z v_x & v_z v_y & v_z^2 \end{pmatrix} \quad (1.37)$$

Perfect Fluid:

$$T^{\mu\nu} = (\rho + p)U^{\mu} U^{\nu} + p\eta^{\mu\nu} \quad (1.38)$$

Under arbitrary boost:

$$T^{\mu'\nu'} = \begin{pmatrix} \gamma^2(\rho + p) - p & -\gamma^2 v_x(\rho + p) & -\gamma^2 v_y(\rho + p) & -\gamma^2 v_z(\rho + p) \\ -\gamma^2 v_x(\rho + p) & \gamma^2 v_x^2(\rho + p) + p & \gamma^2 v_x v_y(\rho + p) & \gamma^2 v_x v_z(\rho + p) \\ -\gamma^2 v_y(\rho + p) & \gamma^2 v_y v_x(\rho + p) & \gamma^2 v_y^2(\rho + p) + p & \gamma^2 v_y v_z(\rho + p) \\ -\gamma^2 v_z(\rho + p) & \gamma^2 v_z v_x(\rho + p) & \gamma^2 v_z v_y(\rho + p) & \gamma^2 v_z^2(\rho + p) + p \end{pmatrix} \quad (1.39)$$

Conservation Law

$$\partial_\mu T^{\mu\nu} = \partial_\mu(\rho + p)U^\mu U^\nu + (\rho + p)U^\mu \partial_\mu U^\nu + (\rho + p)U^\nu \partial_\mu U^\mu + \partial^\nu p = 0 \quad (1.40)$$

Contracting with U_ν yields energy conservation (continuity eq: $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$). Projecting orthogonally to U^μ yields momentum conservation (Euler eq: $\rho[\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v}] = -\nabla p$).

1.10 Classical Field Theory*Euler-Lagrange Equations*

From $S = \int \mathcal{L}(\Phi^i, \partial_\mu \Phi^i) d^4x$:

$$\frac{\partial \mathcal{L}}{\partial \Phi^i} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \Phi^i)} \right) = 0 \quad (1.41)$$

Scalar Field: $\mathcal{L} = -\frac{1}{2}\eta^{\mu\nu}(\partial_\mu \phi)(\partial_\nu \phi) - V(\phi)$ yields Klein-Gordon:

$$\square \phi + m^2 \phi = 0 \quad (1.42)$$

Electromagnetism: $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + A_\mu J^\mu$ yields $\partial_\mu F^{\nu\mu} = J^\nu$.

Derived Energy-Momentum Tensors

$$T^{\mu\nu} * \text{scalar} = (\partial^\mu \phi)(\partial^\nu \phi) - \eta^{\mu\nu} \left[\frac{1}{2}(\partial * \lambda \phi)(\partial^\lambda \phi) + V(\phi) \right] \quad (1.43)$$

$$T^{\mu\nu} * \text{EM} = F^{\mu\lambda} F^\nu{}_\lambda - \frac{1}{4}\eta^{\mu\nu} F^{\lambda\sigma} F_{\lambda\sigma} \quad (1.44)$$

Chapter 2

Manifolds

2.1 Gravity as Geometry

Gravity is a manifestation of spacetime geometry, founded on the **Principle of Equivalence**.

The Weak Equivalence Principle (WEP)

WEP asserts inertial mass (m_i) equals gravitational mass (m_g).

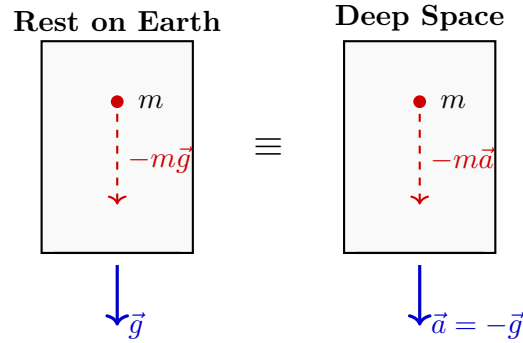
$$F = m_i a \quad (2.1)$$

$$F_g = -m_g \nabla \Phi \quad (2.2)$$

Setting $m_i = m_g$, test particle acceleration is independent of mass:

$$a = -\nabla \Phi \quad (2.3)$$

Locally, a uniform gravitational field is indistinguishable from uniform acceleration.



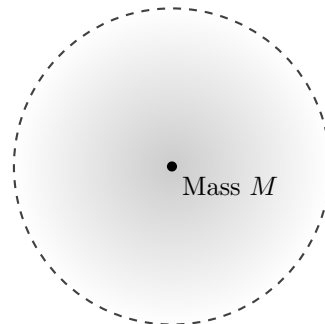
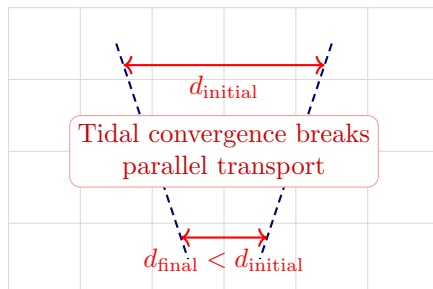
The Einstein (EEP) and Strong (SEP) Equivalence Principles

EEP states no local experiment can distinguish uniform acceleration from a gravitational field. Physics locally reduces to special relativity. SEP extends this to all laws, including gravity itself.

From Equivalence to Geometry

Gravity isn't a force; "unaccelerated" paths are redefined as freely-falling. Global inertial frames don't exist due to tidal forces. We define **locally inertial frames** modeled as a **differentiable manifold**.

Tidal Analysis Perspective



Prediction: Gravitational Redshift

Accelerating rockets in flat spacetime separated by z :

1. Emitted photon wavelength λ_0 travels distance z , taking $\Delta t \approx z/c$.
2. Velocity increases by $\Delta v = a\Delta t = az/c$.
3. Doppler shift yields:

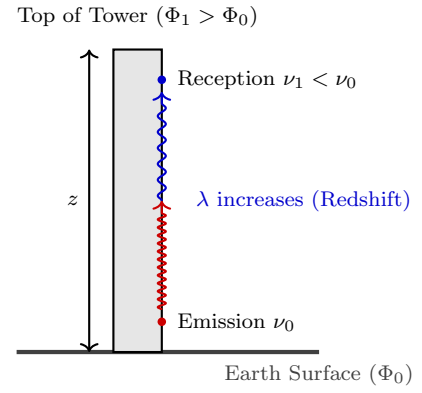
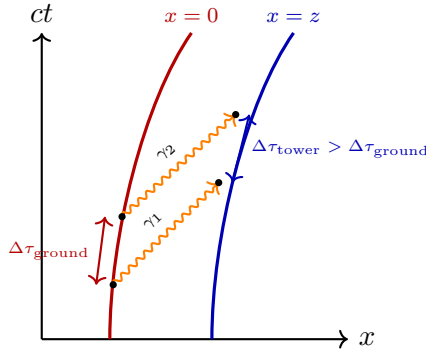
$$\frac{\Delta\lambda}{\lambda_0} = \frac{\lambda_1 - \lambda_0}{\lambda_0} \approx \frac{\Delta v}{c} \quad (2.4)$$

$$\frac{\Delta\lambda}{\lambda_0} = \frac{(az/c)}{c} = \frac{az}{c^2} \quad (2.5)$$

$$\frac{\Delta\lambda}{\lambda_0} = \frac{a_g z}{c^2} \quad (2.6)$$

Using $a_g \approx \Delta\Phi/z$, we get:

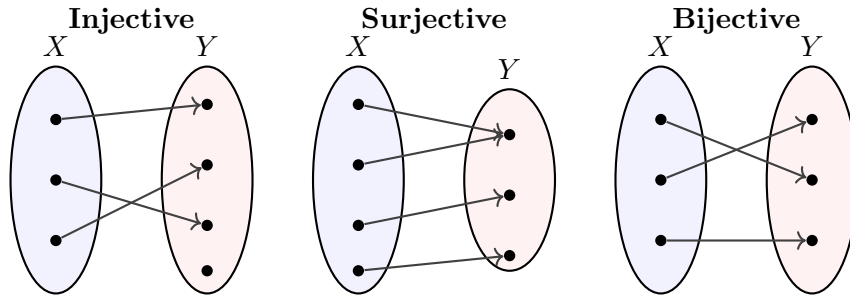
$$\frac{\Delta\lambda}{\lambda_0} \approx \Delta\Phi \quad (2.7)$$

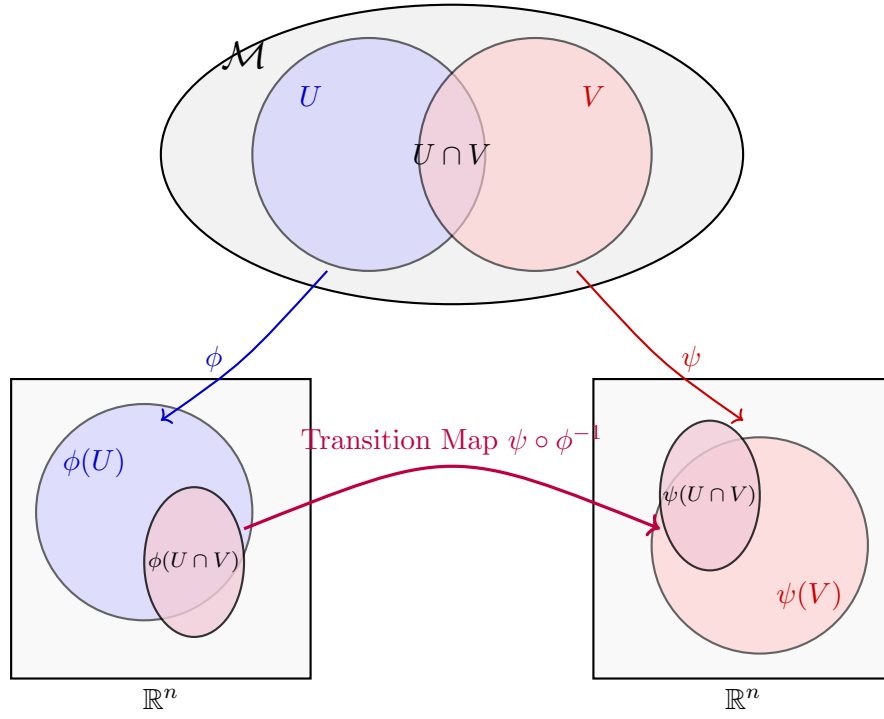


2.2 What Is a Manifold?

A differentiable manifold locally resembles $\mathbb{R}^n$. Examples: $\mathbb{R}^n$, n -sphere S^n , n -torus T^n , Riemann Surfaces, Lie Groups, Direct Products.

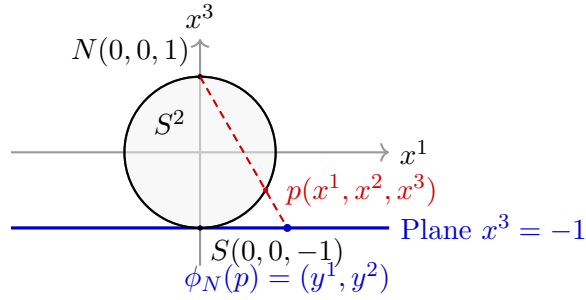
Formal Definition





Stereographic Projection on S^2

Let S^2 be $(x^1)^2 + (x^2)^2 + (x^3)^2 = 1$.



$$(y^1, y^2) = \phi_N(x^1, x^2, x^3) = \left(\frac{2x^1}{1-x^3}, \frac{2x^2}{1-x^3} \right) \quad (2.8)$$

Chart 2 (South Pole S excluded): Projects to plane $x^3 = +1$ with coordinates (z^1, z^2) .

$$(z^1, z^2) = \phi_S(x^1, x^2, x^3) = \left(\frac{2x^1}{1+x^3}, \frac{2x^2}{1+x^3} \right) \quad (2.9)$$

Derivation: Transition Map for S^2

Invert ϕ_N : Let $Y^2 = (y^1)^2 + (y^2)^2$.

$$Y^2 = \frac{4(x^1)^2 + 4(x^2)^2}{(1-x^3)^2} = \frac{4(1-x^3)(1+x^3)}{(1-x^3)^2} = \frac{4(1+x^3)}{1-x^3}$$

$$Y^2(1-x^3) = 4(1+x^3) \implies x^3 = \frac{Y^2 - 4}{Y^2 + 4}$$

$$1 - x^3 = \frac{8}{Y^2 + 4}$$

$$x^1 = \frac{y^1(1-x^3)}{2} = \frac{4y^1}{Y^2+4}, \quad x^2 = \frac{4y^2}{Y^2+4}$$

Apply ϕ_S :

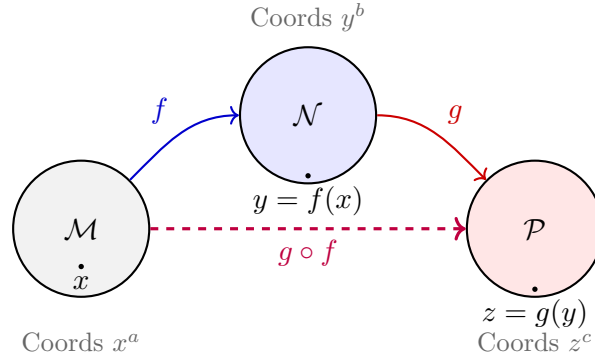
$$(z^1, z^2) = \left(\frac{2x^1}{1+x^3}, \frac{2x^2}{1+x^3} \right)$$

$$1+x^3 = \frac{2Y^2}{Y^2+4}$$

$$z^1 = \frac{4y^1}{(y^1)^2 + (y^2)^2}, \quad z^2 = \frac{4y^2}{(y^1)^2 + (y^2)^2}$$

Transition map is C^∞ except at $(0,0)$ (South Pole, excluded).

The Chain Rule



$$\frac{\partial (g \circ f)^c}{\partial x^a} = \sum_{b=1}^n \frac{\partial f^b}{\partial x^a} \frac{\partial g^c}{\partial y^b} \quad (2.10)$$

$$\frac{\partial}{\partial x^a} = \sum_{b=1}^n \frac{\partial y^b}{\partial x^a} \frac{\partial}{\partial y^b} \quad (2.11)$$

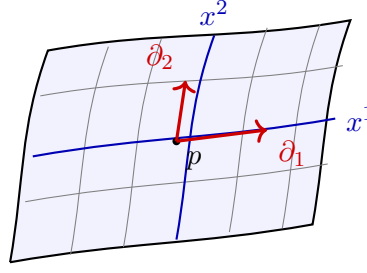
2.3 Vectors Again

T_p is defined as the set of directional derivative operators at p . Linearity: $X = a \frac{d}{d\lambda_1} + b \frac{d}{d\lambda_2}$. Leibniz Rule:

$$\begin{aligned} X(fg) &= \left(a \frac{d}{d\lambda_1} + b \frac{d}{d\lambda_2} \right) (fg) \\ &= f \left(a \frac{dg}{d\lambda_1} + b \frac{dg}{d\lambda_2} \right) + g \left(a \frac{df}{d\lambda_1} + b \frac{df}{d\lambda_2} \right) = f(X(g)) + g(X(f)) \end{aligned}$$

The Coordinate Basis

Partial derivatives $\{\partial_\mu\}$ form a basis.



Derivation:

$$\begin{aligned} \frac{d}{d\lambda}(f) &= \frac{d}{d\lambda}[(f \circ \phi^{-1}) \circ (\phi \circ \gamma)] \\ &= \frac{dx^\mu}{d\lambda} \frac{\partial f}{\partial x^\mu} \\ \frac{d}{d\lambda} &= \frac{dx^\mu}{d\lambda} \partial_\mu \end{aligned} \tag{2.12}$$

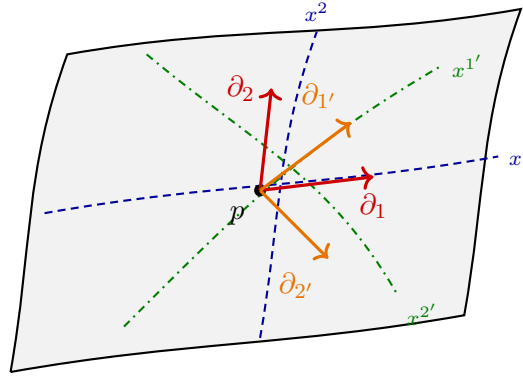
Transformation Properties

Basis Vector Transformation:

$$\partial_{\mu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu \tag{2.13}$$

Vector Component Transformation:

$$\begin{aligned} V &= V^{\mu'} \left(\frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu \right) \implies V^\mu = V^{\mu'} \frac{\partial x^\mu}{\partial x^{\mu'}} \\ V^{\mu'} &= \frac{\partial x^{\mu'}}{\partial x^\mu} V^\mu \end{aligned} \tag{2.14}$$



The Commutator (Lie Bracket)

$$[X, Y](f) \equiv X(Y(f)) - Y(X(f)) \tag{2.15}$$

$$\begin{aligned} [X, Y]^\nu &= X(Y^\mu \partial_\mu x^\nu) - Y(X^\lambda \partial_\lambda x^\nu) \\ &= X(Y^\nu) - Y(X^\nu) \\ &= (X^\lambda \partial_\lambda) Y^\nu - (Y^\mu \partial_\mu) X^\nu \\ [X, Y]^\nu &= X^\lambda \partial_\lambda Y^\nu - Y^\lambda \partial_\lambda X^\nu \end{aligned} \tag{2.16}$$

2.4 Tensors Again

Dual Vectors (One-Forms)

T_p^* maps $T_p \rightarrow \mathbb{R}$.

$$df \left(\frac{d}{d\lambda} \right) = \frac{df}{d\lambda} \quad (2.17)$$

Basis and Components

$\{dx^\mu\}$ forms the dual basis to $\{\partial_\nu\}$.

$$\begin{aligned} dx^\mu(\partial_\nu) &\equiv \partial_\nu(x^\mu) = \frac{\partial x^\mu}{\partial x^\nu} = \delta_\nu^\mu \\ dx^\mu(\partial_\nu) &= \delta_\nu^\mu \end{aligned} \quad (2.18)$$

Transformation Properties

$$dx^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\mu} dx^\mu \quad (2.19)$$

$$\begin{aligned} \omega &= \omega_{\mu'} \left(\frac{\partial x^{\mu'}}{\partial x^\mu} dx^\mu \right) \\ \omega_{\nu'} &= \frac{\partial x^\mu}{\partial x^{\nu'}} \omega_\mu \end{aligned} \quad (2.20)$$

General Tensors

$$T = T^{\mu_1 \dots \mu_k} * \nu_1 \dots \nu_l \partial * \mu_1 \otimes \dots \otimes \partial_{\mu_k} \otimes dx^{\nu_1} \otimes \dots \otimes dx^{\nu_l} \quad (2.21)$$

$$T^{\mu_1 \dots \mu_k} * \nu_1 \dots \nu_l = T(dx^{\mu_1}, \dots, dx^{\mu_k}, \partial * \nu_1, \dots, \partial_{\nu_l}) \quad (2.22)$$

$$T^{\mu'_1 \dots \mu'_k * k} * \nu'_1 \dots \nu'_l = \frac{\partial x^{\mu'_1}}{\partial x^{\mu_1}} \dots \frac{\partial x^{\nu_l}}{\partial x^{\nu'_l}} \dots T^{\mu_1 \dots \mu_k} * \nu_1 \dots \nu_l \quad (2.23)$$

Example: Transforming a (0,2) Tensor

$$S = S_{\mu\nu} dx^\mu dx^\nu = 1, (dx)^2 + x^2(dy)^2 \quad (2.24)$$

Coordinate change: $x' = 2x/y, y' = y/2$. Inverting gives $y = 2y', x = x'y'$. Differentials:

$$dx = y'dx' + x'dy', \quad dy = 2dy'$$

Substitute into S :

$$S = (y'dx' + x'dy')^2 + (x'y')^2(2dy')^2$$

$$S = (y')^2(dx')^2 + x'y'(dx'dy' + dy'dx') + [(x')^2 + 4(x'y')^2](dy')^2$$

$$S_{\mu'\nu'} = \begin{pmatrix} (y')^2 & x'y' \\ x'y' & (x')^2 + 4(x'y')^2 \end{pmatrix} \quad (2.25)$$

Verifying via full transformation law:

$$S_{\mu'\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} S_{\mu\nu} \quad (2.26)$$

$$\begin{aligned}
S_{x'x'} &= (y')(y')(1) = (y')^2 \\
S_{x'y'} &= (y')(x')(1) = x'y' \\
S_{y'y'} &= (x')(x')(1) + (2)(2)(x^2) = (x')^2 + 4(x'y')^2
\end{aligned}$$

The Failure of Partial Derivatives

$\partial_\mu W_\nu$ is not a tensor.

$$\begin{aligned}
\partial_{\mu'} W_{\nu'} &= \frac{\partial}{\partial x^{\mu'}} \left(\frac{\partial x^\nu}{\partial x^{\nu'}} W_\nu \right) \\
\partial_{\mu'} W_{\nu'} &= \left(\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial^2 x^\nu}{\partial x^\mu \partial x^{\nu'}} \right) W_\nu + \frac{\partial x^\nu}{\partial x^{\nu'}} \left(\frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu W_\nu \right) \\
\partial_{\mu'} W_{\nu'} &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} (\partial_\mu W_\nu) + W_\nu \frac{\partial^2 x^\nu}{\partial x^{\mu'} \partial x^{\nu'}} \quad (2.27)
\end{aligned}$$

The second term vanishes only for linear transformations. This motivates the **covariant derivative**.

2.5 The Metric

The metric $g_{\mu\nu}$ is a nondegenerate, symmetric (0,2) tensor.

$$g^{\mu\nu} g_{\nu\sigma} = \delta_\sigma^\mu \quad (2.28)$$

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \quad (2.29)$$

Example: Flat Space in Spherical Coordinates

$x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$ Differentials:

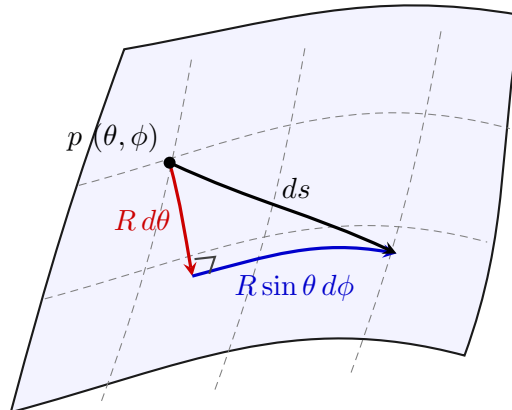
$$\begin{aligned}
dx &= (\sin \theta \cos \phi) dr + (r \cos \theta \cos \phi) d\theta - (r \sin \theta \sin \phi) d\phi \\
dy &= (\sin \theta \sin \phi) dr + (r \cos \theta \sin \phi) d\theta + (r \sin \theta \cos \phi) d\phi \\
dz &= (\cos \theta) dr - (r \sin \theta) d\theta
\end{aligned}$$

Summing $(dx)^2 + (dy)^2 + (dz)^2$:

$$\begin{aligned}
ds^2 &= [\sin^2 \theta dr^2 + r^2 \cos^2 \theta d\theta^2 + r^2 \sin^2 \theta d\phi^2 + 2r \sin \theta \cos \theta dr d\theta] \\
&\quad + [\cos^2 \theta dr^2 - 2r \sin \theta \cos \theta dr d\theta + r^2 \sin^2 \theta d\theta^2] \\
ds^2 &= dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \quad (2.30)
\end{aligned}$$

For the 2-sphere S^2 (constant $r = R$):

$$ds^2 = R^2 d\theta^2 + R^2 \sin^2 \theta d\phi^2 \quad (2.31)$$



Locally Inertial Coordinates (LIC)

In a LIC $x^{\hat{\mu}}$, gravity vanishes to first order:

$$g_{\hat{\mu}\hat{\nu}}(p) = \eta_{\hat{\mu}\hat{\nu}} \quad \partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}}(p) = 0 \quad (2.32)$$

Degrees of freedom (4D):

1. Zeroth Order: 16 parameters to fix 10 components. 2. First Order: 40 parameters to fix 40 components ($\partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}} = 0$). 3. Second Order: 80 parameters to fix 100 components ($\partial_{\hat{\rho}} \partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}} = 0$). Short by 20 parameters, defining the Riemann curvature tensor.

Relative Velocity Example: In observer's LIC, $U^{\hat{\mu}} = (1, 0, 0, 0)$. Rocket $V^{\hat{\mu}} = (\gamma, \gamma v^1, \gamma v^2, \gamma v^3)$.

$$\eta_{\hat{\mu}\hat{\nu}} U^{\hat{\mu}} V^{\hat{\nu}} = -\gamma$$

$$v = \sqrt{1 - \gamma^{-2}} = \sqrt{1 - (\eta_{\hat{\mu}\hat{\nu}} U^{\hat{\mu}} V^{\hat{\nu}})^{-2}}$$

Covariant expression:

$$v = \sqrt{1 - (g_{\mu\nu} U^{\mu} V^{\nu})^{-2}} \quad (2.33)$$

2.6 An Expanding Universe

Robertson-Walker (RW) metric:

$$ds^2 = -dt^2 + a^2(t)[(dx)^2 + (dy)^2 + (dz)^2] \quad (2.34)$$

For $a(t) = t^q$ ($0 < q < 1$):

$$a(t) = t^q \quad (2.35)$$

Light Cones and Particle Horizon

Null path ($ds^2 = 0$):

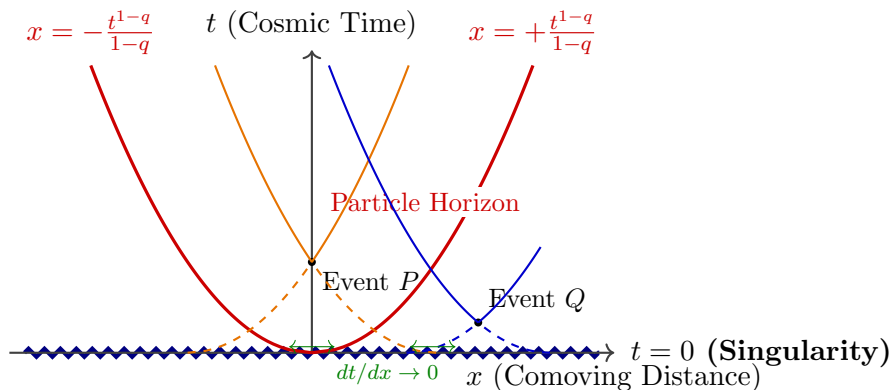
$$0 = -dt^2 + t^{2q}(dx)^2 \implies \frac{dx}{dt} = \pm t^{-q}$$

Justification via tangent vector $V = \frac{dx^{\mu}}{d\lambda} \partial_{\mu}$:

$$\begin{aligned} g(V, V) &= -1 \left(\frac{dt}{d\lambda} \right)^2 + a(t)^2 \left(\frac{dx}{d\lambda} \right)^2 = 0 \\ - \left(\frac{dt}{d\lambda} \right)^2 + t^{2q} \left(\frac{dx}{d\lambda} \right)^2 &= 0 \implies \left(\frac{dx}{dt} \right)^2 = t^{-2q} \end{aligned} \quad (2.36)$$

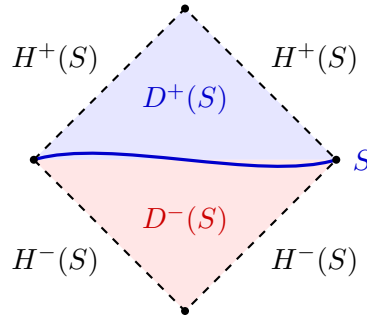
Particle horizon integration:

$$x(t) = \pm \int_{t_0}^t (t')^{-q} dt' = \pm \frac{t^{1-q}}{1-q}$$



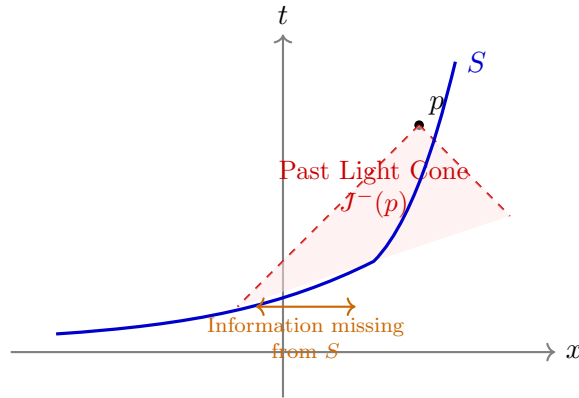
2.7 Causality

Definitions: Causal Curve (timelike/null), Causal Future $J^+(S)$, Chronological Future $I^+(S)$, Achronal Set. Domain of Dependence $D(S)$: state uniquely determined by S . Cauchy Horizon $H(S)$ is the boundary. Cauchy Surface gives $D(\Sigma) = M$ (Globally Hyperbolic).



Failures of Causal Structure

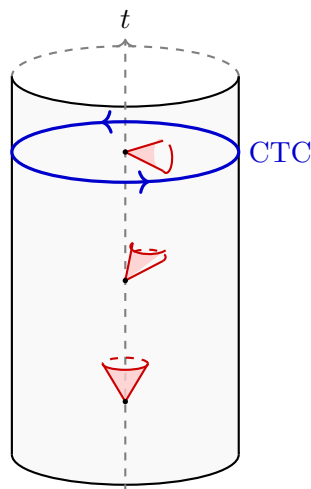
1. Poor Choice of Hypersurface: Slice not a Cauchy surface.



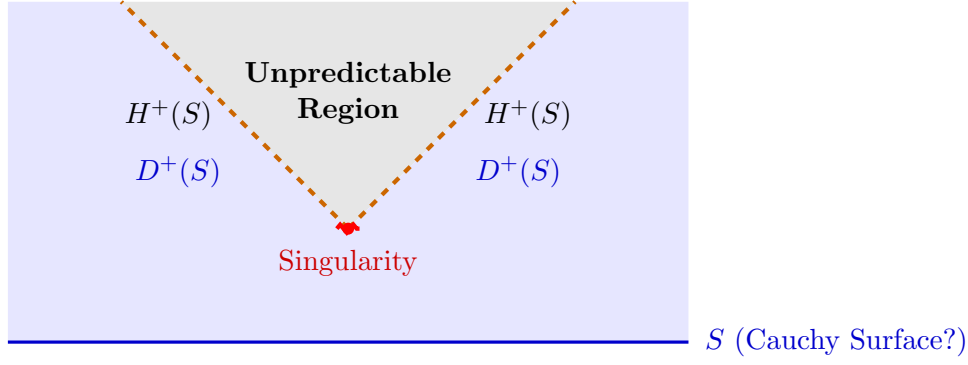
2. Closed Timelike Curves (CTCs): e.g., $R \times S^1$ metric:

$$ds^2 = -\cos(\lambda)dt^2 - \sin(\lambda)(dtdx + dxdt) + \cos(\lambda)dx^2 \quad (2.37)$$

where $\lambda = \cot^{-1}(t)$. Light cones tilt, allowing CTCs.



3. Singularities: Create Cauchy horizons where paths hit an "edge".



2.8 Tensor Densities

Densities transform with Jacobian determinant $|J|^W$.

The Levi-Civita Symbol

$\tilde{\epsilon}_{\mu_1\mu_2\ldots\mu_n}$ is ± 1 or 0 .

$$\begin{aligned}\tilde{\epsilon}_{\mu'_1\ldots\mu'_n}|J|^{-1} &= \tilde{\epsilon}_{\mu_1\ldots\mu_n} \frac{\partial x^{\mu_1}}{\partial x^{\mu'_1}} \cdots \frac{\partial x^{\mu_n}}{\partial x^{\mu'_n}} \\ \tilde{\epsilon}_{\mu'_1\ldots\mu'_n} &= |J| \left(\tilde{\epsilon} * \mu_1 \ldots \mu_n \frac{\partial x^{\mu_1}}{\partial x^{\mu'_1}} \cdots \frac{\partial x^{\mu_n}}{\partial x^{\mu'_n}} \right)\end{aligned}\quad (2.38)$$

Weight $W = +1$.

Determinant of the Metric

$$g' = \det(A^T g A) = (\det(A))^2 g \quad g' = (|J|^{-1})^2 g \implies g' = |J|^{-2} g \quad (2.39)$$

Weight $W = -2$.

The Levi-Civita Tensor

Absorb $\sqrt{|g|}$ ($W = -1$) to make a true tensor ($W = 0$):

$$\epsilon_{\mu_1\mu_2\ldots\mu_n} = \sqrt{|g|} \tilde{\epsilon} * \mu_1\mu_2\ldots\mu_n \quad (2.40)$$

Contravariant tensor:

$$\begin{aligned}\epsilon^{\mu_1\ldots\mu_n} &= \sqrt{|g|} g^{\alpha_1\mu_1} \ldots g^{\alpha_n\mu_n} \tilde{\epsilon} * \alpha_1 \ldots \alpha_n = \sqrt{|g|} |g|^{-1} \tilde{\epsilon}^{\mu_1\ldots\mu_n} \\ \epsilon^{\mu_1\mu_2\ldots\mu_n} &= \frac{1}{\sqrt{|g|}} \tilde{\epsilon}^{\mu_1\mu_2\ldots\mu_n}\end{aligned}\quad (2.41)$$

Contraction Identity

$$\epsilon^{\mu_1\ldots\mu_n} \epsilon_{\nu_1\ldots\nu_n} = (-1)^s, \tilde{\epsilon}^{\mu_1\ldots\mu_n} \tilde{\epsilon} * \nu_1 \ldots \nu_n \quad (2.42)$$

$$\tilde{\epsilon}^{\mu_1\ldots\mu_n} \tilde{\epsilon} * \nu_1 \ldots \nu_n = \delta^{\mu_1\ldots\mu_n} * \nu_1 \ldots \nu_n = n!, \delta^{[\mu_1} * \nu_1 \ldots \delta^{\mu_n]} * \nu_n \quad (2.43)$$

Contracting p indices:

$$\delta^{\mu_1\ldots\mu_p\alpha_1\ldots\alpha_{n-p}} * \mu_1 \ldots \mu_p \beta_1 \ldots \beta_{n-p} = p!, \delta^{\alpha_1\ldots\alpha_{n-p}} * \beta_1 \ldots \beta_{n-p} = p!(n-p)!, \delta^{[\alpha_1} * \beta_1 \ldots \delta^{\alpha_{n-p}]} * \beta_{n-p} \quad (2.44)$$

$$\epsilon^{\mu_1\ldots\mu_p\alpha_1\ldots\alpha_{n-p}} \epsilon_{\mu_1\ldots\mu_p\beta_1\ldots\beta_{n-p}} = (-1)^s p!(n-p)! \delta^{[\alpha_1} * \beta_1 \ldots \delta^{\alpha_{n-p}]} * \beta_{n-p} \quad (2.45)$$

Most common case ($p = n - 1$):

$$\epsilon^{\mu_1\ldots\mu_{n-1}\alpha} \epsilon_{\mu_1\ldots\mu_{n-1}\beta} = (-1)^s (n-1)! \delta^\alpha_\beta \quad (2.46)$$

2.9 Differential Forms

A p -form is a totally antisymmetric $(0, p)$ tensor. Dimension $\binom{n}{p}$.

The Wedge Product

$$(A \wedge B) * \mu_1 \dots \mu * p + q = \frac{(p+q)!}{p!q!} A_{[\mu_1 \dots \mu_p} B_{\mu_{p+1} \dots \mu_{p+q}]} \quad (2.47)$$

$$A \wedge B = (-1)^{pq} B \wedge A \quad (2.48)$$

The Exterior Derivative

d turns a p -form into a $(p+1)$ -form:

$$(dA) * \mu_1 \dots \mu * p + 1 = (p+1) \partial_{[\mu_1} A_{\mu_2 \dots \mu_{p+1}]} \quad (2.49)$$

Graded Leibniz Rule:

$$d(A \wedge B) = (dA) \wedge B + (-1)^p A \wedge (dB) \quad (2.50)$$

dA is a tensor because the non-tensorial part is symmetric and cancels under antisymmetrization:

$$\begin{aligned} (dA) * \mu' \nu' &= \partial * \mu' A_{\nu'} - \partial_{\nu'} A_{\mu'} \\ \text{Bad part} &= W_\nu \frac{\partial^2 x^\nu}{\partial x^{\mu'} \partial x^{\nu'}} - W_\nu \frac{\partial^2 x^\nu}{\partial x^{\mu'} \partial x^{\nu'}} = 0 \end{aligned}$$

$d^2 = 0$ (exact forms are closed):

$$d(dA) = 0 \quad (2.51)$$

The Double Dual Identity

$$\star \star A = (-1)^{s+p(n-p)} A \quad (2.52)$$

$$(\star \star A) * \rho = \frac{1}{(n-p)!} \epsilon^{\sigma \dots} * \rho \dots \left(\frac{1}{p!} \epsilon^{\nu \dots} * \sigma \dots A * \nu \dots \right)$$

Maxwell's Equations

Faraday 2-form:

$$F = dA \quad (2.53)$$

Homogeneous:

$$dF = d(dA) = 0 \quad (2.54)$$

Inhomogeneous:

$$d(\star F) = \star J \quad (2.55)$$

Tensor form: $\star d \star F = \star \star J = J \implies \nabla_\mu F^{\mu\nu} = J^\nu$.

$$\nabla_\mu F^{\mu\nu} = J^\nu \quad (2.56)$$

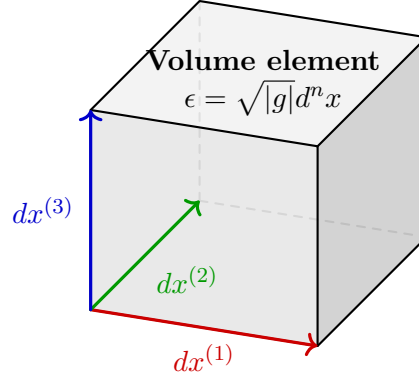
2.10 Integration

Volume element transforms as density $W = +1$:

$$d^n x' = \left| \frac{\partial x^{\mu'}}{\partial x^\mu} \right| d^n x = |J| d^n x \quad (2.57)$$

Integration maps an n -form ω to $\mathbb{R}$:

$$\int_{\Sigma} : \omega \rightarrow \mathbb{R} \quad (2.58)$$



The Invariant Volume Element

Combining $\sqrt{|g|}$ ($W = -1$) and $d^n x$ ($W = +1$) yields invariant n -form.

$$\epsilon = \frac{1}{n!} \sqrt{|g|} \tilde{\epsilon}_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_n}$$

Using $\frac{1}{n!} \tilde{\epsilon}_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_n} = d^n x$:

$$\epsilon = \sqrt{|g|} d^n x \quad (2.59)$$

Invariant integral:

$$S = \int \phi(x) \sqrt{|g|} d^n x \quad (2.60)$$

$$S = \int \phi(x) \epsilon \quad (2.61)$$

Chapter 3

Curvature

3.1 Overview

Curvature measures deviations from flat space, formalized by the Riemann tensor. This is built via a hierarchy of structures.

1. **The Connection (Christoffel Symbols):** Connects tangent spaces at different points.

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2}g^{\lambda\sigma}(\partial_{\mu}g_{\nu\sigma} + \partial_{\nu}g_{\sigma\mu} - \partial_{\sigma}g_{\mu\nu}). \quad (3.1)$$

2. **The Covariant Derivative:** Upgrades partial derivatives to transform as tensors.

$$\nabla_{\mu}V^{\nu} = \partial_{\mu}V^{\nu} + \Gamma_{\mu\sigma}^{\nu}V^{\sigma}. \quad (3.2)$$

3. **Geodesics:** Paths of zero acceleration in curved spacetime.

$$\frac{d^2x^{\mu}}{d\lambda^2} + \Gamma_{\rho\sigma}^{\mu}\frac{dx^{\rho}}{d\lambda}\frac{dx^{\sigma}}{d\lambda} = 0. \quad (3.3)$$

4. **The Riemann Curvature Tensor:** The definitive measure of true curvature.

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu}\Gamma_{\nu\sigma}^{\rho} - \partial_{\nu}\Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho}\Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho}\Gamma_{\mu\sigma}^{\lambda}. \quad (3.4)$$

3.2 Covariant Derivatives

Partial derivatives of tensors fail to transform properly due to coordinate grid curvature:

$$V^{\nu'} = \frac{\partial x^{\nu'}}{\partial x^{\nu}}V^{\nu}. \quad (3.5)$$

$$\partial_{\mu'}V^{\nu'} = \underbrace{\frac{\partial x^{\mu}}{\partial x^{\mu'}}\frac{\partial x^{\nu'}}{\partial x^{\nu}}(\partial_{\mu}V^{\nu})}_{\text{Tensor part}} + \underbrace{\frac{\partial x^{\mu}}{\partial x^{\mu'}}\frac{\partial^2 x^{\nu'}}{\partial x^{\mu}\partial x^{\nu}}V^{\nu}}_{\text{Non-tensorial part}}. \quad (3.6)$$

Defining the Covariant Derivative

The covariant derivative ∇_{μ} corrects this by incorporating the connection Γ :

$$\nabla_{\mu}V^{\nu} = \partial_{\mu}V^{\nu} + \Gamma_{\mu\lambda}^{\nu}V^{\lambda}. \quad (3.7)$$

Requiring $\nabla_{\mu'}V^{\nu'} = \frac{\partial x^{\mu}}{\partial x^{\mu'}}\frac{\partial x^{\nu'}}{\partial x^{\nu}}\nabla_{\mu}V^{\nu}$ enforces the transformation law for Γ :

$$\Gamma^{\nu'}{}_{\mu'\lambda'} = \frac{\partial x^{\mu}}{\partial x^{\mu'}}\frac{\partial x^{\lambda}}{\partial x^{\lambda'}}\frac{\partial x^{\nu'}}{\partial x^{\nu}}\Gamma^{\nu}{}_{\mu\lambda} - \frac{\partial x^{\mu}}{\partial x^{\mu'}}\frac{\partial x^{\lambda}}{\partial x^{\lambda'}}\frac{\partial^2 x^{\nu'}}{\partial x^{\mu}\partial x^{\lambda}}. \quad (3.8)$$

For a one-form ω_{λ} , preserving the Leibniz rule yields a negative connection term:

$$\nabla_{\mu}\omega_{\lambda} = \partial_{\mu}\omega_{\lambda} - \Gamma_{\mu\lambda}^{\sigma}\omega_{\sigma}. \quad (3.9)$$

The Christoffel Connection

General relativity relies on the unique metric-compatible ($\nabla_\rho g_{\mu\nu} = 0$) and torsion-free ($\Gamma_{\mu\nu}^\lambda = \Gamma_{\nu\mu}^\lambda$) connection.

1. Covariant Derivative of the Inverse Metric:

$$\nabla_\rho(g^{\mu\sigma}g_{\sigma\nu}) = 0 \implies (\nabla_\rho g^{\mu\sigma})g_{\sigma\nu} = 0 \implies \nabla_\rho g^{\mu\lambda} = 0. \quad (3.10)$$

2. Covariant Derivative of the Levi-Civita Tensor: In locally inertial coordinates ($\Gamma = 0, \partial g = 0$), $\partial_\lambda g = 0 \implies \partial_\lambda \sqrt{|g|} = 0$.

$$\nabla_\lambda \epsilon_{\mu\nu\rho\sigma} = \partial_\lambda(\sqrt{|g|}\tilde{\epsilon}_{\mu\nu\rho\sigma}) = 0. \quad (3.11)$$

Permuting metric compatibility $\nabla_\rho g_{\mu\nu} = 0$ yields the Christoffel formula:

$$\Gamma_{\mu\nu}^\sigma = \frac{1}{2}g^{\sigma\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}). \quad (3.12)$$

Divergence Formula

Contracting indices on the Christoffel symbol provides an identity for the metric determinant g :

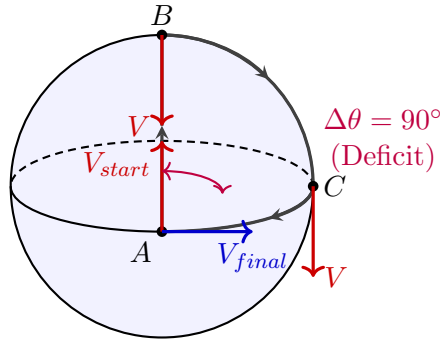
$$\Gamma_{\mu\lambda}^\mu = \frac{1}{2}g^{\mu\nu}\partial_\lambda g_{\mu\nu} = \frac{1}{2}\left(\frac{1}{g}\partial_\lambda g\right) = \frac{1}{\sqrt{|g|}}\partial_\lambda \sqrt{|g|}. \quad (3.13)$$

This leads to the divergence formula and Stokes' theorem in curved space:

$$\nabla_\mu V^\mu = \frac{1}{\sqrt{|g|}}\partial_\mu(\sqrt{|g|}V^\mu) \quad (3.14)$$

$$\int_{\mathcal{V}} d^n x \sqrt{|g|} \nabla_\mu V^\mu = \int_{\partial\mathcal{V}} d^{n-1}x \sqrt{|\gamma|} n_\mu V^\mu \quad (3.15)$$

3.3 Parallel Transport and Geodesics



A vector V^μ is parallel transported along $x^\mu(\lambda)$ if its directional covariant derivative vanishes:

$$\frac{DV^\mu}{d\lambda} \equiv \frac{dx^\nu}{d\lambda} \nabla_\nu V^\mu = 0 \implies \frac{dV^\mu}{d\lambda} + \Gamma_{\sigma\rho}^\mu \frac{dx^\sigma}{d\lambda} V^\rho = 0. \quad (3.16)$$

Geodesic Equation

A geodesic parallel transports its own tangent vector $U^\mu = dx^\mu/d\lambda$:

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0. \quad (3.17)$$

Variational Principle

Geodesics also extremize the energy integral $I = \frac{1}{2} \int g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu d\tau$. Varying $\delta I = 0$ yields:

$$g_{\sigma\nu} \ddot{x}^\nu + \frac{1}{2} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu = 0. \quad (3.18)$$

Multiplying by $g^{\lambda\sigma}$ recovers the geodesic equation exactly.

Null Geodesics

Massless particles follow null geodesics ($ds^2 = 0$) parameterized by affine parameter λ such that $p^\mu = dx^\mu/d\lambda$:

$$p^\nu \nabla_\nu p^\mu = 0. \quad (3.19)$$

Observed energy is $E = -p_\mu U_{\text{obs}}^\mu$.

3.4 Properties of Geodesics*Affine Parameters*

Affine parameters λ are unique up to linear transformations $\alpha = a\lambda + b$. Under an arbitrary parameter change $\alpha(\lambda)$:

$$\left(\frac{d\alpha}{d\lambda}\right)^2 \left[\frac{d^2 x^\mu}{d\alpha^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\alpha} \frac{dx^\sigma}{d\alpha} \right] + \frac{dx^\mu}{d\alpha} \frac{d^2 \alpha}{d\lambda^2} = 0. \quad (3.20)$$

For α to be affine, $d^2 \alpha/d\lambda^2 = 0$.

Riemann Normal Coordinates (RNC)

The exponential map allows construction of RNC $x^{\hat{\mu}}(q) = k^{\hat{\mu}}$, where geodesics from p are straight lines $x^{\hat{\mu}}(\lambda) = \lambda k^{\hat{\mu}}$. At p :

$$\Gamma^{\hat{\mu}} * \hat{\rho} \hat{\sigma}(p) = 0 \implies \partial * \hat{\rho} g_{\hat{\mu}\hat{\nu}}(p) = 0. \quad (3.21)$$

3.5 The Expanding Universe Revisited

Robertson-Walker metric ($k = 0$):

$$ds^2 = -dt^2 + a^2(t) \delta_{ij} dx^i dx^j. \quad (3.22)$$

Varying $I = \frac{1}{2} \int [-(\dot{t})^2 + a^2(t) \delta_{ij} \dot{x}^i \dot{x}^j] d\tau$ gives connection coefficients:

$$\Gamma_{00}^0 = 0, \quad \Gamma_{0i}^0 = 0, \quad \Gamma_{ij}^0 = a\dot{a}\delta_{ij}, \quad (3.23)$$

$$\Gamma_{0j}^k = \frac{\dot{a}}{a} \delta_j^k, \quad \Gamma_{ij}^k = 0, \quad \Gamma_{00}^k = 0. \quad (3.24)$$

Cosmological Redshift

For a null geodesic $\frac{dx}{dt} = \frac{1}{a(t)}$, the t -equation implies:

$$\frac{d}{d\lambda} \ln \left(\frac{dt}{d\lambda} \right) = -\frac{d}{d\lambda} \ln a \implies \frac{dt}{d\lambda} = \frac{\omega_0}{a(t)}. \quad (3.25)$$

Energy $E = -p_\mu U^\mu \propto 1/a(t)$.

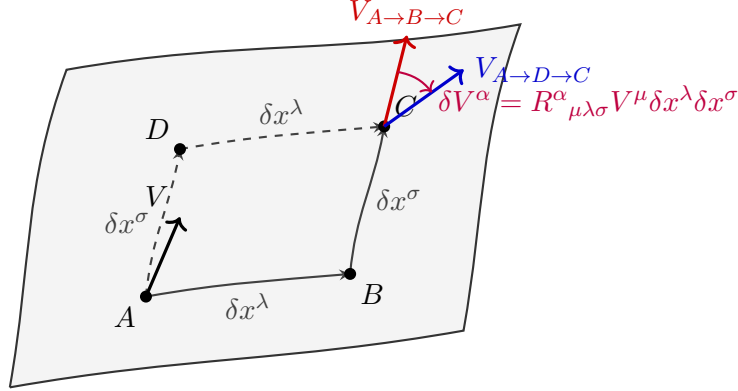
Conservation of Energy-Momentum

For perfect fluid $T^{\mu\nu} = (\rho + p)U^\mu U^\nu + pg^{\mu\nu}$, $\nabla_\mu T^{\mu 0} = 0$ implies:

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + p) = 0. \quad (3.26)$$

Given $p = w\rho$, $\rho \propto a^{-3(1+w)}$.

3.6 The Riemann Curvature Tensor



The commutator of covariant derivatives acting on vector V^ρ reveals curvature:

$$[\nabla_\mu, \nabla_\nu]V^\rho = \left(\partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda \right) V^\sigma - 2\Gamma_{[\mu\nu]}^\lambda \nabla_\lambda V^\rho. \quad (3.27)$$

For a torsion-free connection, this defines the Riemann tensor $R^\rho_{\sigma\mu\nu}$:

$$[\nabla_\mu, \nabla_\nu]V^\rho = R^\rho_{\sigma\mu\nu} V^\sigma. \quad (3.28)$$

Change in V^α around a small loop $\delta x^\lambda, \delta x^\sigma$:

$$\delta V^\alpha = R^\alpha_{\mu\lambda\sigma} V^\mu \delta x^\lambda \delta x^\sigma. \quad (3.29)$$

Theorem: Flatness

$R^\rho_{\sigma\mu\nu} = 0$ everywhere if and only if there exists a coordinate system where $\partial_\sigma g_{\mu\nu} = 0$ everywhere. Proof involves defining a parallel-transported dual basis $\nabla_\mu \omega_\nu = 0$, implying $d\omega = 0 \implies \omega_a = dy^a$, yielding constant metric $ds^2 = \eta_{ab} dy^a dy^b$.

3.7 Properties of the Riemann Tensor

Symmetries

Evaluated in RNC, $R_{\rho\sigma\mu\nu} = \frac{1}{2}(\partial_\mu \partial_\sigma g_{\rho\nu} - \partial_\mu \partial_\rho g_{\nu\sigma} - \partial_\nu \partial_\sigma g_{\rho\mu} + \partial_\nu \partial_\rho g_{\mu\sigma})$ yields:

$$R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu} \quad (\text{Antisymmetry first pair}) \quad (3.30)$$

$$R_{\rho\sigma\mu\nu} = -R_{\rho\sigma\nu\mu} \quad (\text{Antisymmetry last pair}) \quad (3.31)$$

$$R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma} \quad (\text{Symmetry pair exchange}) \quad (3.32)$$

$$R_{\rho[\sigma\mu\nu]} = 0 \quad (\text{First Bianchi Identity}) \quad (3.33)$$

These reduce independent components to $\frac{1}{12}n^2(n^2 - 1)$ (20 components in 4D).

Second Bianchi Identity

Differentiating R in RNC gives the differential identity:

$$\nabla_{[\lambda} R_{\rho\sigma]\mu\nu} = 0. \quad (3.34)$$

Ricci, Einstein, and Weyl Tensors

Ricci Tensor / Scalar:

$$R_{\mu\nu} = R^\lambda{}_\mu \lambda_\nu, \quad R = g^{\mu\nu} R_{\mu\nu}. \quad (3.35)$$

Einstein Tensor: Contracting the Second Bianchi Identity twice yields the divergence-free Einstein tensor:

$$\nabla^\mu \left(R_{\rho\mu} - \frac{1}{2} R g_{\rho\mu} \right) = 0 \implies G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}, \quad \nabla^\mu G_{\mu\nu} = 0. \quad (3.36)$$

Weyl Tensor (Trace-Free):

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu} - \frac{2}{n-2} (g_{\rho[\mu} R_{\nu]\sigma} - g_{\sigma[\mu} R_{\nu]\rho}) + \frac{2}{(n-1)(n-2)} g_{\rho[\mu} g_{\nu]\sigma} R. \quad (3.37)$$

3.8 Mathematical Interlude: Diffeomorphisms and Lie Derivatives

The Lie Derivative $\mathcal{L}_V$ measures tensor change along vector field V^μ flow without a connection:

$$\mathcal{L} * V U^\mu = [V, U]^\mu = V^\nu \partial_\nu U^\mu - U^\nu \partial_\nu V^\mu \quad (3.38)$$

$$\mathcal{L} * V \omega * \mu = V^\nu \partial_\nu \omega_\mu + \omega_\nu \partial_\mu V^\nu \quad (3.39)$$

$$\mathcal{L} * V T^\mu * \nu = V^\lambda \nabla_\lambda T^\mu * \nu - (\nabla * \lambda V^\mu) T^\lambda * \nu + (\nabla * \nu V^\lambda) T^\mu{}_\lambda \quad (3.40)$$

3.9 Symmetries and Killing Vectors

A metric symmetry $\partial_{\sigma*} g_{\mu\nu} = 0$ implies conserved momentum along geodesics $dp_{\sigma*}/d\tau = 0$. For generator K^μ , $K^\nu p_\nu$ is conserved.

$$\frac{d}{d\tau} (K_\nu p^\nu) = p^\mu p^\nu \nabla_{(\mu} K_{\nu)} = 0. \quad (3.41)$$

This gives **Killing's Equation**:

$$\nabla_{(\mu} K_{\nu)} = \frac{1}{2} (\nabla_\mu K_\nu + \nabla_\nu K_\mu) = 0. \quad (3.42)$$

Using the covariant derivative expansion, $\nabla_\mu K_\nu = \partial_\mu K_\nu - \Gamma_{\mu\nu}^\lambda K_\lambda$.

Chapter 4

Gravitation

4.1 *Physics in Curved Spacetime*

In Newtonian gravity, body response and potential generation are governed by:

$$\mathbf{a} = -\nabla\Phi \quad \implies \quad \frac{d^2x^i}{dt^2} = -\partial_i\Phi. \quad (4.1)$$

$$\nabla^2\Phi = 4\pi G\rho, \quad (4.2)$$

The Minimal-Coupling Principle

Based on the Einstein Equivalence Principle (EEP), laws generalize to curved spacetime via the minimal-coupling principle (Comma-Goes-to-Semicolon Rule: $\eta_{\mu\nu} \rightarrow g_{\mu\nu}$, $\partial_\mu \rightarrow \nabla_\mu$).

Derivation of the Geodesic Equation

In flat spacetime, a free particle path $x^\mu(\lambda)$ obeys:

$$\frac{d^2x^\mu}{d\lambda^2} = 0. \quad (4.3)$$

Rewriting using the directional derivative:

$$\frac{d}{d\lambda} = \frac{dx^\nu}{d\lambda} \partial_\nu. \quad (4.4)$$

$$\frac{dx^\nu}{d\lambda} \partial_\nu \left(\frac{dx^\mu}{d\lambda} \right) = 0. \quad (4.5)$$

Applying minimal coupling ($\partial_\nu \rightarrow \nabla_\nu$):

$$\frac{dx^\nu}{d\lambda} \nabla_\nu \left(\frac{dx^\mu}{d\lambda} \right) = 0. \quad (4.6)$$

Expanding the covariant derivative:

$$\nabla_\nu V^\mu = \partial_\nu V^\mu + \Gamma^\mu_{\nu\rho} V^\rho. \quad (4.7)$$

Substituting back:

$$\begin{aligned} \frac{dx^\nu}{d\lambda} \left[\partial_\nu \left(\frac{dx^\mu}{d\lambda} \right) + \Gamma^\mu_{\nu\rho} \frac{dx^\rho}{d\lambda} \right] &= 0 \\ \frac{dx^\nu}{d\lambda} \partial_\nu \left(\frac{dx^\mu}{d\lambda} \right) + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{d\lambda} \frac{dx^\rho}{d\lambda} &= 0. \end{aligned} \quad (4.8)$$

Yielding the **geodesic equation**:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\nu\rho} \frac{dx^\nu}{d\lambda} \frac{dx^\rho}{d\lambda} = 0, \quad (4.9)$$

Energy-Momentum Conservation

Flat spacetime conservation $\partial_\mu T^{\mu\nu} = 0$ generalizes to:

$$\nabla_\mu T^{\mu\nu} = 0. \quad (4.10)$$

The Newtonian Limit

Requires: 1. Slow motion ($v \ll c$), 2. Static field ($\partial_0 g_{\alpha\beta} = 0$), 3. Weak field ($g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$).

1. Simplifying the Geodesic Equation

Slow motion implies:

$$\left| \frac{dx^i}{d\tau} \right| \ll \left| \frac{dt}{d\tau} \right|. \quad (4.11)$$

Geodesic equation simplifies to:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{00}^\mu \left(\frac{dt}{d\tau} \right)^2 = 0. \quad (4.12)$$

2. Evaluating the Christoffel Symbol

$$\Gamma^\mu * 00 = \frac{1}{2} g^{\mu\lambda} (\partial * 0 g_{\lambda 0} + \partial_0 g_{0\lambda} - \partial_\lambda g_{00}). \quad (4.13)$$

Static field assumption gives:

$$\Gamma^\mu * 00 = -\frac{1}{2} g^{\mu\lambda} \partial * \lambda g_{00}. \quad (4.14)$$

3. The Weak Field Approximation

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \quad (4.15)$$

Inverse metric to linear order $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$:

$$\begin{aligned} g^{\mu\lambda} g_{\lambda\nu} &= (\eta^{\mu\lambda} - h^{\mu\lambda})(\eta_{\lambda\nu} + h_{\lambda\nu}) \\ &= \eta^{\mu\lambda} \eta_{\lambda\nu} + \eta^{\mu\lambda} h_{\lambda\nu} - h^{\mu\lambda} \eta_{\lambda\nu} - h^{\mu\lambda} h_{\lambda\nu} \\ &= \delta^\mu * \nu + \eta^{\mu\lambda} h * \lambda_\nu - (\eta^{\mu\rho} \eta^{\lambda\sigma} h_{\rho\sigma}) \eta_{\lambda\nu} + \mathcal{O}(h^2) \\ &= \delta^\mu * \nu + \eta^{\mu\lambda} h * \lambda_\nu - \eta^{\mu\rho} \delta^\sigma * \nu h * \rho\sigma \\ &= \delta^\mu * \nu + \eta^{\mu\lambda} h * \lambda_\nu - \eta^{\mu\rho} h_{\rho\nu}. \end{aligned} \quad (4.16)$$

Replacing inverse metric with flat metric:

$$\Gamma^\mu * 00 \approx -\frac{1}{2} \eta^{\mu\lambda} \partial * \lambda h_{00}. \quad (4.17)$$

Substituting into (4.12):

$$\frac{d^2 x^\mu}{d\tau^2} = \frac{1}{2} \eta^{\mu\lambda} \partial_\lambda h_{00} \left(\frac{dt}{d\tau} \right)^2. \quad (4.18)$$

4. Decomposing Time and Spatial Components

Time Component ($\mu = 0$):

$$\frac{d^2 t}{d\tau^2} = \frac{1}{2} \eta^{0\lambda} \partial_\lambda h_{00} \left(\frac{dt}{d\tau} \right)^2 = \frac{1}{2} \eta^{00} \partial_0 h_{00} \left(\frac{dt}{d\tau} \right)^2. \quad (4.19)$$

Static field gives:

$$\frac{d^2 t}{d\tau^2} = 0 \quad \implies \quad \frac{dt}{d\tau} = \text{constant}. \quad (4.20)$$

Spatial Components ($\mu = i$):

$$\frac{d^2 x^i}{d\tau^2} = \frac{1}{2} \eta^{ij} \partial_j h_{00} \left(\frac{dt}{d\tau} \right)^2 = \frac{1}{2} \delta^{ij} \partial_j h_{00} \left(\frac{dt}{d\tau} \right)^2 = \frac{1}{2} \partial_i h_{00} \left(\frac{dt}{d\tau} \right)^2. \quad (4.21)$$

Applying chain rule:

$$\frac{d}{d\tau} = \frac{dt}{d\tau} \frac{d}{dt} \quad \Longrightarrow \quad \frac{d^2 x^i}{d\tau^2} = \frac{d}{d\tau} \left(\frac{dt}{d\tau} \frac{dx^i}{dt} \right). \quad (4.22)$$

$$\frac{d^2 x^i}{d\tau^2} = \left(\frac{d^2 t}{d\tau^2} \right) \frac{dx^i}{dt} + \frac{dt}{d\tau} \frac{d}{d\tau} \left(\frac{dx^i}{dt} \right). \quad (4.23)$$

$$\frac{d^2 x^i}{d\tau^2} = \left(\frac{dt}{d\tau} \right)^2 \frac{d^2 x^i}{dt^2}. \quad (4.24)$$

Substituting back:

$$\left(\frac{dt}{d\tau} \right)^2 \frac{d^2 x^i}{dt^2} = \frac{1}{2} \partial_i h_{00} \left(\frac{dt}{d\tau} \right)^2 \quad \Longrightarrow \quad \frac{d^2 x^i}{dt^2} = \frac{1}{2} \partial_i h_{00}. \quad (4.25)$$

5. Connection to the Newtonian Potential

Equating to Newtonian acceleration (4.1):

$$\frac{d^2 x^i}{dt^2} = \frac{1}{2} \partial_i h_{00} \quad \text{vs} \quad \frac{d^2 x^i}{dt^2} = -\partial_i \Phi. \quad (4.26)$$

$$\frac{1}{2} \partial_i h_{00} = -\partial_i \Phi \quad \Longrightarrow \quad h_{00} = -2\Phi. \quad (4.27)$$

$$g_{00} = -(1 + 2\Phi). \quad (4.28)$$

4.2 Einstein's Equation

4.2.1 Motivating the Tensorial Equation

Generalizing Poisson's equation:

$$\nabla^2 \Phi = 4\pi G \rho, \quad (4.29)$$

$$[\nabla^2 g] * \mu\nu \propto T * \mu\nu. \quad (4.30)$$

Naive guess:

$$R_{\mu\nu} = \kappa T_{\mu\nu}, \quad (4.31)$$

4.2.2 The Conservation of Energy Problem

Local energy conservation $\nabla^\mu T_{\mu\nu} = 0$ implies $\nabla^\mu R_{\mu\nu} = 0$. However, the contracted Bianchi identity states:

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu R, \quad (4.32)$$

$$g^{\mu\nu} R_{\mu\nu} = \kappa g^{\mu\nu} T_{\mu\nu} \quad \Longrightarrow \quad R = \kappa T. \quad (4.33)$$

$$\nabla^\mu R_{\mu\nu} = \frac{1}{2} \nabla_\nu (\kappa T). \quad (4.34)$$

$$\frac{1}{2} \kappa \nabla_\nu T = 0 \quad \Longrightarrow \quad \nabla_\nu T = 0. \quad (4.35)$$

4.2.3 The Einstein Tensor

Rearranging the contracted Bianchi identity:

$$\nabla^\mu R_{\mu\nu} - \frac{1}{2} \nabla_\nu R = 0. \quad (4.36)$$

$$\nabla^\mu R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \nabla^\mu R = 0 \quad \implies \quad \nabla^\mu \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) = 0. \quad (4.37)$$

Defining the **Einstein Tensor**:

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}. \quad (4.38)$$

Proposed field equation:

$$G_{\mu\nu} = \kappa T_{\mu\nu}. \quad (4.39)$$

4.2.4 The Trace-Reversed Form

Taking the trace in 4D:

$$\begin{aligned} g^{\mu\nu} G_{\mu\nu} &= \kappa g^{\mu\nu} T_{\mu\nu} \\ g^{\mu\nu} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) &= \kappa T \\ R - \frac{1}{2} R (g^{\mu\nu} g_{\mu\nu}) &= \kappa T. \end{aligned} \quad (4.40)$$

$$R - \frac{1}{2} R(4) = \kappa T \quad \implies \quad R - 2R = \kappa T \quad \implies \quad R = -\kappa T. \quad (4.41)$$

$$R_{\mu\nu} - \frac{1}{2} (-\kappa T) g_{\mu\nu} = \kappa T_{\mu\nu}. \quad (4.42)$$

Yielding the **trace-reversed form**:

$$R_{\mu\nu} = \kappa \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right). \quad (4.43)$$

4.2.5 The Newtonian Limit and the Coupling Constant

For pressureless dust:

$$T_{\mu\nu} = \rho U_\mu U_\nu. \quad (4.44)$$

Using weak-field metric $g_{00} = -1 + h_{00}$:

$$g_{00}(U^0)^2 = -1 \quad \implies \quad (-1 + h_{00})(U^0)^2 = -1 \quad \implies \quad (U^0)^2 = (1 - h_{00})^{-1}. \quad (4.45)$$

$$U^0 \approx 1 + \frac{1}{2} h_{00}. \quad (4.46)$$

$$U_0 = (-1 + h_{00}) \left(1 + \frac{1}{2} h_{00} \right) \approx -1 - \frac{1}{2} h_{00} + h_{00} = -1 + \frac{1}{2} h_{00}. \quad (4.47)$$

To leading order:

$$T_{00} = \rho (U_0)^2 = \rho (-1)^2 = \rho. \quad (4.48)$$

$$T = g^{00} T_{00} \approx (-1) \rho = -\rho. \quad (4.49)$$

Trace-reversed 00-component:

$$\begin{aligned} R_{00} &= \kappa \left(T_{00} - \frac{1}{2} T g_{00} \right) \\ R_{00} &\approx \kappa \left[\rho - \frac{1}{2} (-\rho) (-1) \right] \\ R_{00} &\approx \kappa \left(\rho - \frac{1}{2} \rho \right) = \frac{1}{2} \kappa \rho. \end{aligned} \quad (4.50)$$

Evaluating the Ricci Component R_{00}

$$R_{00} = R^\lambda * 0\lambda 0 = R^0 * 000 + R^i * 0i0. \quad (4.51)$$

$$R^i * 0j0 = \partial_j \Gamma^i * 00 - \partial * 0\Gamma^i * j0 + \Gamma^i * j\lambda \Gamma^\lambda * 00 - \Gamma^i * 0\lambda \Gamma^\lambda * j0. \quad (4.52)$$

Static, weak-field limits:

$$R^i * 0j0 \approx \partial_j \Gamma^i * 00. \quad (4.53)$$

$$\begin{aligned} R * 00 &= R^i * 0i0 = \partial * i(\Gamma^i * 00) \\ &= \partial * i \left(-\frac{1}{2} \delta^{ij} \partial_j h_{00} \right) \\ &= -\frac{1}{2} \delta^{ij} \partial_i \partial_j h_{00} = -\frac{1}{2} \nabla^2 h_{00}. \end{aligned} \quad (4.54)$$

Extracting Newton's Constant

$$-\frac{1}{2} \nabla^2 h_{00} = \frac{1}{2} \kappa \rho \quad \implies \quad \nabla^2 h_{00} = -\kappa \rho. \quad (4.55)$$

Using $h_{00} = -2\Phi$:

$$\nabla^2(-2\Phi) = -\kappa \rho \quad \implies \quad 2\nabla^2 \Phi = \kappa \rho \quad \implies \quad \nabla^2 \Phi = \frac{1}{2} \kappa \rho. \quad (4.56)$$

Matching $\nabla^2 \Phi = 4\pi G \rho$ yields $\kappa = 8\pi G$. **Einstein's Field Equation:**

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (4.57)$$

4.2.6 Vacuum Equations

$$R_{\mu\nu} = 0. \quad (4.58)$$

Lagrangian Formulation in Curved Spacetime

2. Integration by Parts in Curved Space

$$\nabla_\mu (AB) = (\nabla_\mu A)B + A(\nabla_\mu B) \quad (4.59)$$

$$(\nabla_\mu A)B = \nabla_\mu (AB) - A(\nabla_\mu B) \quad (4.60)$$

$$\int (\nabla_\mu A)B \sqrt{-g}, d^n x = \int \nabla_\mu (AB) \sqrt{-g}, d^n x - \int A(\nabla_\mu B) \sqrt{-g}, d^n x \quad (4.61)$$

$$\int (\nabla_\mu A)B \sqrt{-g}, d^n x = - \int A(\nabla_\mu B) \sqrt{-g}, d^n x \quad (4.62)$$

3. Deriving the Euler-Lagrange Equations

$$\delta S = \int \left[\frac{\partial \hat{\mathcal{L}}}{\partial \Phi} \delta \Phi + \frac{\partial \hat{\mathcal{L}}}{\partial (\nabla_\mu \Phi)} \delta (\nabla_\mu \Phi) \right] \sqrt{-g}, d^n x = 0 \quad (4.63)$$

$$\delta S = \int \left[\frac{\partial \hat{\mathcal{L}}}{\partial \Phi} \delta \Phi + \frac{\partial \hat{\mathcal{L}}}{\partial (\nabla_\mu \Phi)} \nabla_\mu (\delta \Phi) \right] \sqrt{-g}, d^n x = 0 \quad (4.64)$$

$$\delta S = \int \left[\frac{\partial \hat{\mathcal{L}}}{\partial \Phi} \delta \Phi - \nabla_\mu \left(\frac{\partial \hat{\mathcal{L}}}{\partial (\nabla_\mu \Phi)} \right) \delta \Phi \right] \sqrt{-g}, d^n x = 0 \quad (4.65)$$

$$\delta S = \int \left[\frac{\partial \hat{\mathcal{L}}}{\partial \Phi} - \nabla_\mu \left(\frac{\partial \hat{\mathcal{L}}}{\partial (\nabla_\mu \Phi)} \right) \right] \delta \Phi \sqrt{-g}, d^n x = 0 \quad (4.66)$$

4. Expanding the Scalar Field Example

$$\hat{\mathcal{L}} = -\frac{1}{2}g^{\alpha\beta}(\nabla_\alpha\phi)(\nabla_\beta\phi) - V(\phi) \quad (4.67)$$

$$\frac{\partial\hat{\mathcal{L}}}{\partial\phi} = -\frac{dV}{d\phi} \quad (4.68)$$

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -\frac{1}{2}g^{\alpha\beta} \left[\frac{\partial(\nabla_\alpha\phi)}{\partial(\nabla_\mu\phi)}(\nabla_\beta\phi) + (\nabla_\alpha\phi)\frac{\partial(\nabla_\beta\phi)}{\partial(\nabla_\mu\phi)} \right] \quad (4.69)$$

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -\frac{1}{2}g^{\alpha\beta} \left[\delta_\alpha^\mu(\nabla_\beta\phi) + (\nabla_\alpha\phi)\delta_\beta^\mu \right] \quad (4.70)$$

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -\frac{1}{2} \left[g^{\mu\beta}\nabla_\beta\phi + g^{\alpha\mu}\nabla_\alpha\phi \right] \quad (4.71)$$

$$\frac{\partial\hat{\mathcal{L}}}{\partial(\nabla_\mu\phi)} = -g^{\mu\nu}\nabla_\nu\phi \quad (4.72)$$

$$-\frac{dV}{d\phi} - \nabla_\mu(-g^{\mu\nu}\nabla_\nu\phi) = 0 \quad (4.73)$$

$$-\frac{dV}{d\phi} + g^{\mu\nu}\nabla_\mu\nabla_\nu\phi = 0 \quad (4.74)$$

$$\square\phi - \frac{dV}{d\phi} = 0 \quad (4.75)$$

4.2.7 The Hilbert Action

$$S_H = \int \sqrt{-g}R, d^n x. \quad (4.76)$$

4.2.8 Varying the Hilbert Action

$$(\delta g^{\mu\lambda})g_{\lambda\nu} + g^{\mu\lambda}(\delta g_{\lambda\nu}) = 0. \quad (4.77)$$

$$\delta g_{\mu\nu} = -g_{\mu\rho}g_{\nu\sigma}\delta g^{\rho\sigma}. \quad (4.78)$$

$$\begin{aligned} \delta S_H &= \delta \int d^n x \sqrt{-g}(g^{\mu\nu}R_{\mu\nu}) \\ &= \int d^n x \left[\sqrt{-g}g^{\mu\nu}(\delta R_{\mu\nu}) + \sqrt{-g}R_{\mu\nu}(\delta g^{\mu\nu}) + R(g^{\mu\nu}R_{\mu\nu})(\delta\sqrt{-g}) \right] \\ &\equiv (\delta S) * 1 + (\delta S) * 2 + (\delta S)_3. \end{aligned} \quad (4.79)$$

Evaluating $(\delta S)_1$: Variation of the Ricci Tensor

$$R^\rho * \mu\lambda\nu = \partial * \lambda\Gamma^\rho * \nu\mu - \partial * \nu\Gamma^\rho * \lambda\mu + \Gamma^\rho * \lambda\sigma\Gamma^\sigma * \nu\mu - \Gamma^\rho * \nu\sigma\Gamma^\sigma * \lambda\mu. \quad (4.80)$$

$$\nabla * \lambda(\delta\Gamma^\rho * \nu\mu) = \partial * \lambda(\delta\Gamma^\rho * \nu\mu) + \Gamma^\rho * \lambda\sigma\delta\Gamma^\sigma * \nu\mu - \Gamma^\sigma * \lambda\nu\delta\Gamma^\rho * \sigma\mu - \Gamma^\sigma * \lambda\mu\delta\Gamma^\rho * \nu\sigma. \quad (4.81)$$

Palatini identity:

$$\delta R^\rho * \mu\lambda\nu = \nabla_\lambda(\delta\Gamma^\rho * \nu\mu) - \nabla * \nu(\delta\Gamma^\rho * \lambda\mu). \quad (4.82)$$

$$\delta R * \mu\nu = \nabla_\rho(\delta\Gamma^\rho * \nu\mu) - \nabla * \nu(\delta\Gamma^\rho * \rho\mu). \quad (4.83)$$

$$(\delta S) * 1 = \int d^n x \sqrt{-g}g^{\mu\nu} [\nabla_\rho(\delta\Gamma^\rho * \nu\mu) - \nabla * \nu(\delta\Gamma^\rho * \rho\mu)]. \quad (4.84)$$

$$(\delta S) * 1 = \int d^n x \sqrt{-g}\nabla_\sigma \left[g^{\mu\nu}(\delta\Gamma^\sigma * \mu\nu) - g^{\mu\sigma}(\delta\Gamma^\lambda * \lambda\mu) \right] \equiv \int d^n x \sqrt{-g}\nabla_\sigma V^\sigma. \quad (4.85)$$

$$\delta\Gamma^\sigma * \mu\nu = \frac{1}{2}g^{\sigma\rho}(\nabla * \mu\delta g_{\nu\rho} + \nabla_\nu\delta g_{\mu\rho} - \nabla_\rho\delta g_{\mu\nu}). \quad (4.86)$$

$$g^{\mu\nu}\delta\Gamma^\sigma * \mu\nu = \nabla * \mu(\delta g^{\sigma\mu}) - \nabla^\sigma(\delta g^\mu * \mu), \quad (4.87)$$

$$g^{\mu\sigma}\delta\Gamma^\lambda * \lambda\mu = \nabla^\sigma(\delta g^\mu * \mu). \quad (4.88)$$

$$V^\sigma = \nabla * \mu(\delta g^{\sigma\mu}) - \nabla^\sigma(\delta g^\mu * \mu). \quad (4.89)$$

$(\delta S) * 1 = 0$ via Stokes's theorem.

4.3 The Palatini Formulation and Identity

4.3.1 Proof the Palatini Identity

$$R^\rho * \sigma\mu\nu = \partial * \mu\Gamma^\rho * \nu\sigma - \partial * \nu\Gamma^\rho * \mu\sigma + \Gamma^\rho * \mu\lambda\Gamma^\lambda * \nu\sigma - \Gamma^\rho * \nu\lambda\Gamma^\lambda * \mu\sigma. \quad (4.90)$$

$$\Gamma^{\rho'} * \mu'\nu' = \frac{\partial x^{\rho'}}{\partial x^\rho} \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \Gamma^\rho * \mu\nu + \frac{\partial x^{\rho'}}{\partial x^\lambda} \frac{\partial^2 x^\lambda}{\partial x^{\mu'} \partial x^{\nu'}}. \quad (4.91)$$

$$\delta R^\rho * \sigma\mu\nu = \partial_\mu(\delta\Gamma^\rho * \nu\sigma) - \partial * \nu(\delta\Gamma^\rho * \mu\sigma) + (\delta\Gamma^\rho * \mu\lambda)\Gamma^\lambda * \nu\sigma + \Gamma^\rho * \mu\lambda(\delta\Gamma^\lambda * \nu\sigma) - (\delta\Gamma^\rho * \nu\lambda)\Gamma^\lambda * \mu\sigma - \Gamma^\rho * \nu\lambda(\delta\Gamma^\lambda * \mu\sigma). \quad (4.92)$$

$$\begin{aligned} \nabla * \mu(\delta\Gamma^\rho * \nu\sigma) - \nabla * \nu(\delta\Gamma^\rho * \mu\sigma) &= \left[\partial * \mu(\delta\Gamma^\rho * \nu\sigma) + \Gamma^\rho * \mu\lambda\delta\Gamma^\lambda * \nu\sigma - \Gamma^\lambda * \mu\nu\delta\Gamma^\rho * \lambda\sigma - \Gamma^\lambda * \mu\sigma\delta\Gamma^\rho * \nu\lambda \right] \\ &\quad - \left[\partial * \nu(\delta\Gamma^\rho * \mu\sigma) + \Gamma^\rho * \nu\lambda\delta\Gamma^\lambda * \mu\sigma - \Gamma^\lambda * \nu\mu\delta\Gamma^\rho * \lambda\sigma - \Gamma^\lambda * \nu\sigma\delta\Gamma^\rho * \mu\lambda \right] \end{aligned} \quad (4.93)$$

$$\begin{aligned} \nabla * \mu(\delta\Gamma^\rho * \nu\sigma) - \nabla * \nu(\delta\Gamma^\rho * \mu\sigma) &= \partial * \mu(\delta\Gamma^\rho * \nu\sigma) - \partial * \nu(\delta\Gamma^\rho * \mu\sigma) \\ &\quad + \Gamma^\rho * \mu\lambda\delta\Gamma^\lambda * \nu\sigma - \Gamma^\rho * \nu\lambda\delta\Gamma^\lambda * \mu\sigma \\ &\quad - \Gamma^\lambda * \mu\nu\delta\Gamma^\rho * \lambda\sigma + \Gamma^\lambda * \nu\mu\delta\Gamma^\rho * \lambda\sigma \\ &\quad - \Gamma^\lambda * \mu\sigma\delta\Gamma^\rho * \nu\lambda + \Gamma^\lambda * \nu\sigma\delta\Gamma^\rho * \mu\lambda. \end{aligned} \quad (4.94)$$

$$\nabla * \mu(\delta\Gamma^\rho * \nu\sigma) - \nabla * \nu(\delta\Gamma^\rho * \mu\sigma) = \partial * \mu(\delta\Gamma^\rho * \nu\sigma) - \partial * \nu(\delta\Gamma^\rho * \mu\sigma) + \Gamma^\rho * \mu\lambda\delta\Gamma^\lambda * \nu\sigma - \Gamma^\rho * \nu\lambda\delta\Gamma^\lambda * \mu\sigma + \Gamma^\lambda * \nu\sigma\delta\Gamma^\rho * \mu\lambda - \Gamma^\lambda * \mu\sigma\delta\Gamma^\rho * \nu\lambda. \quad (4.95)$$

$$\delta R^\rho * \sigma\mu\nu = \nabla_\mu(\delta\Gamma^\rho * \nu\sigma) - \nabla * \nu(\delta\Gamma^\rho * \mu\sigma). \quad (4.96)$$

$$\delta R * \mu\nu = \nabla_\lambda(\delta\Gamma^\lambda * \mu\nu) - \nabla * \nu(\delta\Gamma^\lambda_{\mu\lambda}). \quad (4.97)$$

4.3.2 Variation of the Action and Integration by Parts

$$\delta_\Gamma S = \int d^4x \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu}. \quad (4.98)$$

$$\delta_\Gamma S = \int d^4x \sqrt{-g} g^{\mu\nu} \left[\nabla_\lambda(\delta\Gamma^\lambda * \mu\nu) - \nabla * \nu(\delta\Gamma^\lambda * \mu\lambda) \right]. \quad (4.99)$$

$$\nabla * \lambda(\mathbf{g}^{\mu\nu} \delta\Gamma^\lambda * \mu\nu) = (\nabla * \lambda \mathbf{g}^{\mu\nu}) \delta\Gamma^\lambda * \mu\nu + \mathbf{g}^{\mu\nu} \nabla * \lambda(\delta\Gamma^\lambda * \mu\nu). \quad (4.100)$$

$$\mathbf{g}^{\mu\nu} \nabla * \lambda(\delta\Gamma^\lambda * \mu\nu) = \nabla * \lambda(\mathbf{g}^{\mu\nu} \delta\Gamma^\lambda * \mu\nu) - (\nabla * \lambda \mathbf{g}^{\mu\nu}) \delta\Gamma^\lambda * \mu\nu. \quad (4.101)$$

$$\mathbf{g}^{\mu\nu} \nabla * \nu(\delta\Gamma^\lambda * \mu\lambda) = \nabla * \nu(\mathbf{g}^{\mu\nu} \delta\Gamma^\lambda * \mu\lambda) - (\nabla * \nu \mathbf{g}^{\mu\nu}) \delta\Gamma^\lambda * \mu\lambda. \quad (4.102)$$

$$\delta * \Gamma S = \int d^4x \left[\nabla_\lambda(\mathbf{g}^{\mu\nu} \delta\Gamma^\lambda * \mu\nu) - \nabla * \nu(\mathbf{g}^{\mu\nu} \delta\Gamma^\lambda * \mu\lambda) - (\nabla * \lambda \mathbf{g}^{\mu\nu}) \delta\Gamma^\lambda * \mu\nu + (\nabla * \nu \mathbf{g}^{\mu\nu}) \delta\Gamma^\lambda * \mu\lambda \right]. \quad (4.103)$$

$$\delta * \Gamma S = \int d^4x \left[-(\nabla_\lambda \mathbf{g}^{\mu\nu}) \delta\Gamma^\lambda * \mu\nu + (\nabla * \nu \mathbf{g}^{\mu\nu}) \delta\Gamma^\lambda * \mu\lambda \right] = 0. \quad (4.104)$$

$$\delta * \Gamma S = \int d^4x \left[-\nabla_\lambda \mathbf{g}^{\mu\nu} + \delta^\nu * \lambda \nabla * \sigma \mathbf{g}^{\mu\sigma} \right] \delta\Gamma^\lambda * \mu\nu = 0. \quad (4.105)$$

$$-\nabla * \lambda \mathbf{g}^{\mu\nu} + \frac{1}{2} \delta^\nu * \lambda \nabla * \sigma \mathbf{g}^{\mu\sigma} + \frac{1}{2} \delta^\mu * \lambda \nabla * \sigma \mathbf{g}^{\nu\sigma} = 0. \quad (4.106)$$

4.3.3 Recovering Metric Compatibility

$$\begin{aligned}
-\nabla_\nu \mathbf{g}^{\mu\nu} + \frac{1}{2}(4)\nabla_\sigma \mathbf{g}^{\mu\sigma} + \frac{1}{2}\delta^\mu * \nu \nabla * \sigma \mathbf{g}^{\nu\sigma} &= 0 \\
-\nabla_\sigma \mathbf{g}^{\mu\sigma} + 2\nabla_\sigma \mathbf{g}^{\mu\sigma} + \frac{1}{2}\nabla_\sigma \mathbf{g}^{\mu\sigma} &= 0 \\
\frac{3}{2}\nabla_\sigma \mathbf{g}^{\mu\sigma} &= 0.
\end{aligned} \tag{4.107}$$

$$\nabla_\lambda \mathbf{g}^{\mu\nu} = 0 \implies \nabla_\lambda (\sqrt{-g} g^{\mu\nu}) = 0. \tag{4.108}$$

$$\sqrt{-g} \nabla_\lambda g^{\mu\nu} + g^{\mu\nu} \nabla_\lambda \sqrt{-g} = 0. \tag{4.109}$$

$$\det(\mathbf{g}^{\mu\nu}) = (\sqrt{-g})^4 \det(g^{\mu\nu}) = g^2 (g^{-1}) = g. \tag{4.110}$$

$$\nabla_\lambda \sqrt{-g} = 0. \tag{4.111}$$

$$\sqrt{-g} \nabla_\lambda g^{\mu\nu} = 0 \implies \nabla_\lambda g^{\mu\nu} = 0. \tag{4.112}$$

Evaluating $(\delta S)_3$: Variation of the Determinant

$$\ln(\det M) = \text{Tr}(\ln M). \tag{4.113}$$

$$\frac{1}{\det M} \delta(\det M) = \text{Tr}(M^{-1} \delta M). \tag{4.114}$$

$$\frac{1}{g} \delta g = \text{Tr}(g^{\mu\nu} \delta g_{\mu\nu}) = g^{\mu\nu} \delta g_{\mu\nu}. \tag{4.115}$$

$$\delta g = g(g^{\mu\nu} \delta g_{\mu\nu}) = g(g^{\mu\nu} [-g_{\mu\rho} g_{\nu\sigma} \delta g^{\rho\sigma}]) = -g(\delta^\nu * \rho g * \nu \sigma \delta g^{\rho\sigma}) = -g(g_{\rho\sigma} \delta g^{\rho\sigma}). \tag{4.116}$$

$$\begin{aligned}
\delta \sqrt{-g} &= \frac{1}{2\sqrt{-g}} (-\delta g) \\
&= \frac{1}{2\sqrt{-g}} (g(g_{\mu\nu} \delta g^{\mu\nu})) \\
&= -\frac{1}{2} \left(\frac{-g}{\sqrt{-g}} \right) g_{\mu\nu} \delta g^{\mu\nu} \\
&= -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}.
\end{aligned} \tag{4.117}$$

$$(\delta S) * 3 = \int d^n x R(\delta \sqrt{-g}) = \int d^n x \sqrt{-g} \left(-\frac{1}{2} R g * \mu\nu \right) \delta g^{\mu\nu}. \tag{4.118}$$

4.3.4 Deriving the Vacuum Equations

$$\begin{aligned}
\delta S_H &= \int d^n x \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} + \int d^n x \sqrt{-g} \left(-\frac{1}{2} R g_{\mu\nu} \right) \delta g^{\mu\nu} \\
\delta S_H &= \int d^n x \sqrt{-g} \left[R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right] \delta g^{\mu\nu}.
\end{aligned} \tag{4.119}$$

$$\delta S = \int \sum_i \left(\frac{\delta S}{\delta \Phi^i} \delta \Phi^i \right) d^n x. \tag{4.120}$$

$$\frac{1}{\sqrt{-g}} \frac{\delta S_H}{\delta g^{\mu\nu}} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}. \tag{4.121}$$

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0. \tag{4.122}$$

4.3.5 Including Matter: The Energy-Momentum Tensor

$$S = \frac{1}{16\pi G} S_H + S_M, \quad (4.123)$$

$$\delta S = \frac{1}{16\pi G} \delta S_H + \delta S_M = 0. \quad (4.124)$$

$$\frac{1}{\sqrt{-g}} \frac{\delta S}{\delta g^{\mu\nu}} = \frac{1}{16\pi G} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) + \frac{1}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}} = 0. \quad (4.125)$$

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = -16\pi G \frac{1}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}}. \quad (4.126)$$

$$T_{\mu\nu} \equiv \frac{-2}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}}. \quad (4.127)$$

Verification via the Scalar Field

$$\begin{aligned} \delta S_\phi &= \delta \int d^n x \left[\sqrt{-g} \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) \right] \\ &= \int d^n x \left[\sqrt{-g} \delta \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) + \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) \delta \sqrt{-g} \right]. \end{aligned} \quad (4.128)$$

$$\delta S_\phi = \int d^n x \sqrt{-g} \delta g^{\mu\nu} \left[-\frac{1}{2} \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \left(-\frac{1}{2} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - V(\phi) \right) \right]. \quad (4.129)$$

$$\begin{aligned} T^{(\phi)} * \mu\nu &= \frac{-2}{\sqrt{-g}} \frac{\delta S * \phi}{\delta g^{\mu\nu}} \\ &= -2 \left[-\frac{1}{2} \nabla_\mu \phi \nabla_\nu \phi + \frac{1}{4} g_{\mu\nu} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi + \frac{1}{2} g_{\mu\nu} V(\phi) \right] \\ T^{(\phi)} * \mu\nu &= \nabla * \mu\phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} g^{\rho\sigma} \nabla_\rho \phi \nabla_\sigma \phi - g_{\mu\nu} V(\phi). \end{aligned} \quad (4.130)$$

The Hilbert vs. Canonical Tensor

$$S^{\mu\nu} = \frac{\delta \mathcal{L}}{\delta(\partial_\mu \Phi^i)} \partial^\nu \Phi^i - \eta^{\mu\nu} \mathcal{L}. \quad (4.131)$$

4.4 Properties of Einstein's Equation

$$\frac{d\theta}{d\tau} = 2\omega^2 - 2\sigma^2 - \frac{1}{3}\theta^2 - R_{\mu\nu} U^\mu U^\nu, \quad (4.132)$$

$$\frac{d\theta}{d\tau} = -R_{\hat{0}\hat{0}}. \quad (4.133)$$

$$R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2} T g_{\mu\nu} \right). \quad (4.134)$$

$$T = \eta^{\hat{\alpha}\hat{\beta}} T_{\hat{\alpha}\hat{\beta}} = -T_{\hat{0}\hat{0}} + T_{\hat{1}\hat{1}} + T_{\hat{2}\hat{2}} + T_{\hat{3}\hat{3}} = -\rho + p_x + p_y + p_z. \quad (4.135)$$

$$\begin{aligned} R_{\hat{0}\hat{0}} &= 8\pi G \left[\rho - \frac{1}{2} (-\rho + p_x + p_y + p_z) (-1) \right] \\ R_{\hat{0}\hat{0}} &= 8\pi G \left[\rho - \frac{1}{2} \rho + \frac{1}{2} (p_x + p_y + p_z) \right] \\ R_{\hat{0}\hat{0}} &= 4\pi G (\rho + p_x + p_y + p_z). \end{aligned} \quad (4.136)$$

$$\frac{d\theta}{d\tau} = -4\pi G (\rho + p_x + p_y + p_z). \quad (4.137)$$

4.4.1 Derivation of the Weyl Tensor Propagation Equation

1. The Contracted Bianchi Identity

$$\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\mu R_{\rho\sigma\nu\lambda} + \nabla_\nu R_{\rho\sigma\lambda\mu} = 0. \quad (4.138)$$

$$\nabla^\rho R_{\rho\sigma\mu\nu} + \nabla_\mu R^\rho * \sigma\nu\rho + \nabla * \nu R^\rho * \sigma\rho\mu = 0. \quad (4.139)$$

$$\nabla * \mu R^\rho * \sigma\nu\rho = -\nabla * \mu R_{\sigma\nu} \quad (4.140)$$

$$\nabla_\nu R^\rho * \sigma\rho\mu = \nabla * \nu R_{\sigma\mu}. \quad (4.141)$$

$$\nabla^\rho R_{\rho\sigma\mu\nu} - \nabla_\mu R_{\sigma\nu} + \nabla_\nu R_{\sigma\mu} = 0 \implies \nabla^\rho R_{\rho\sigma\mu\nu} = \nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}. \quad (4.142)$$

$$\nabla^\rho R_{\rho\sigma\mu\nu} = 2\nabla_{[\mu} R_{\nu]\sigma}. \quad (4.143)$$

2. The Divergence of the Weyl Tensor

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu} - \frac{1}{2}(g_{\rho\mu}R_{\sigma\nu} - g_{\rho\nu}R_{\sigma\mu} - g_{\sigma\mu}R_{\rho\nu} + g_{\sigma\nu}R_{\rho\mu}) + \frac{1}{6}R(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}). \quad (4.144)$$

$$\nabla^\rho C_{\rho\sigma\mu\nu} = \nabla^\rho R_{\rho\sigma\mu\nu} - \frac{1}{2}(\delta^\rho * \mu \nabla^\rho R * \sigma\nu - \delta^\rho * \nu \nabla^\rho R * \sigma\mu - g_{\sigma\mu} \nabla^\rho R_{\rho\nu} + g_{\sigma\nu} \nabla^\rho R_{\rho\mu}) + \frac{1}{6}(\delta^\rho * \mu g * \sigma\nu \nabla^\rho R - \delta^\rho * \nu g * \sigma\mu \nabla^\rho R). \quad (4.145)$$

$$\nabla^\rho C_{\rho\sigma\mu\nu} = (\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) - \frac{1}{2}\left(\nabla_\mu R_{\sigma\nu} - \nabla_\nu R_{\sigma\mu} - \frac{1}{2}g_{\sigma\mu} \nabla_\nu R + \frac{1}{2}g_{\sigma\nu} \nabla_\mu R\right) + \frac{1}{6}(g_{\sigma\nu} \nabla_\mu R - g_{\sigma\mu} \nabla_\nu R). \quad (4.146)$$

$$\begin{aligned} \nabla^\rho C_{\rho\sigma\mu\nu} &= \left(1 - \frac{1}{2}\right)(\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) + \left(\frac{1}{4} - \frac{1}{6}\right)g_{\sigma\mu} \nabla_\nu R - \left(\frac{1}{4} - \frac{1}{6}\right)g_{\sigma\nu} \nabla_\mu R \\ &= \frac{1}{2}(\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) + \frac{1}{12}(g_{\sigma\mu} \nabla_\nu R - g_{\sigma\nu} \nabla_\mu R). \end{aligned} \quad (4.147)$$

$$\frac{1}{2}(\nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma}) = \nabla_{[\mu} R_{\nu]\sigma} \quad (4.148)$$

$$\frac{1}{12}(g_{\sigma\mu} \nabla_\nu R - g_{\sigma\nu} \nabla_\mu R) = \frac{1}{6}\left[\frac{1}{2}(g_{\sigma\mu} \nabla_\nu R - g_{\sigma\nu} \nabla_\mu R)\right] = \frac{1}{6}g_{\sigma[\mu} \nabla_{\nu]} R. \quad (4.149)$$

$$\nabla^\rho C_{\rho\sigma\mu\nu} = \nabla_{[\mu} R_{\nu]\sigma} + \frac{1}{6}g_{\sigma[\mu} \nabla_{\nu]} R. \quad (4.150)$$

3. Introducing the Matter Source

$$R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu}\right). \quad (4.151)$$

$$\begin{aligned} \nabla_\mu R_{\nu\sigma} - \nabla_\nu R_{\mu\sigma} &= 8\pi G \left[\nabla_\mu \left(T_{\nu\sigma} - \frac{1}{2}Tg_{\nu\sigma}\right) - \nabla_\nu \left(T_{\mu\sigma} - \frac{1}{2}Tg_{\mu\sigma}\right) \right] \\ &= 8\pi G \left[(\nabla_\mu T_{\nu\sigma} - \nabla_\nu T_{\mu\sigma}) - \frac{1}{2}(g_{\nu\sigma} \nabla_\mu T - g_{\mu\sigma} \nabla_\nu T) \right]. \end{aligned} \quad (4.152)$$

$$2\nabla_{[\mu} R_{\nu]\sigma} = 8\pi G (2\nabla_{[\mu} T_{\nu]\sigma} - g_{\sigma[\nu} \nabla_{\mu]} T). \quad (4.153)$$

$$\nabla_{[\mu} R_{\nu]\sigma} = 8\pi G \left(\nabla_{[\mu} T_{\nu]\sigma} + \frac{1}{2}g_{\sigma[\mu} \nabla_{\nu]} T \right). \quad (4.154)$$

$$\frac{1}{6}g_{\sigma[\mu} \nabla_{\nu]} R = \frac{1}{6}g_{\sigma[\mu} \nabla_{\nu]} (-8\pi GT) = 8\pi G \left(-\frac{1}{6}g_{\sigma[\mu} \nabla_{\nu]} T \right). \quad (4.155)$$

$$\begin{aligned} \nabla^\rho C_{\rho\sigma\mu\nu} &= 8\pi G \left[\nabla_{[\mu} T_{\nu]\sigma} + \left(\frac{1}{2} - \frac{1}{6}\right)g_{\sigma[\mu} \nabla_{\nu]} T \right] \\ \nabla^\rho C_{\rho\sigma\mu\nu} &= 8\pi G \left(\nabla_{[\mu} T_{\nu]\sigma} + \frac{1}{3}g_{\sigma[\mu} \nabla_{\nu]} T \right). \end{aligned} \quad (4.156)$$

4.4.2 Quantum Gravity and the Planck Scale

$$\text{Planck Mass: } m_P = \left(\frac{\hbar c}{G}\right)^{1/2} \approx 1.22 \times 10^{19}, \text{ GeV} \quad (4.157)$$

$$\text{Planck Length: } l_P = \left(\frac{\hbar G}{c^3}\right)^{1/2} \approx 1.62 \times 10^{-33}, \text{ cm} \quad (4.158)$$

$$\text{Planck Time: } t_P = \left(\frac{\hbar G}{c^5}\right)^{1/2} \approx 5.39 \times 10^{-44}, \text{ sec} \quad (4.159)$$

$$\text{Planck Energy: } E_P = \left(\frac{\hbar c^5}{G}\right)^{1/2} \approx 1.22 \times 10^{19}, \text{ GeV} \quad (4.160)$$

$$[c] = LT^{-1} \quad (4.161)$$

$$[\hbar] = ML^2T^{-1} \quad (4.162)$$

$$[G] = M^{-1}L^3T^{-2} \quad (4.163)$$

$$[m_P] = \left[\frac{\hbar c}{G}\right]^{1/2} = \left(\frac{(ML^2T^{-1})(LT^{-1})}{M^{-1}L^3T^{-2}}\right)^{1/2} = \left(\frac{ML^3T^{-2}}{M^{-1}L^3T^{-2}}\right)^{1/2} = (M^2)^{1/2} = M \quad (4.164)$$

$$[l_P] = \left[\frac{\hbar G}{c^3}\right]^{1/2} = \left(\frac{(ML^2T^{-1})(M^{-1}L^3T^{-2})}{(LT^{-1})^3}\right)^{1/2} = \left(\frac{L^5T^{-3}}{L^3T^{-3}}\right)^{1/2} = (L^2)^{1/2} = L \quad (4.165)$$

$$[t_P] = \left[\frac{\hbar G}{c^5}\right]^{1/2} = \left(\frac{(ML^2T^{-1})(M^{-1}L^3T^{-2})}{(LT^{-1})^5}\right)^{1/2} = \left(\frac{L^5T^{-3}}{L^5T^{-5}}\right)^{1/2} = (T^2)^{1/2} = T \quad (4.166)$$

$$[E_P] = \left[\frac{\hbar c^5}{G}\right]^{1/2} = \left(\frac{(ML^2T^{-1})(L^5T^{-5})}{M^{-1}L^3T^{-2}}\right)^{1/2} = \left(\frac{ML^7T^{-6}}{M^{-1}L^3T^{-2}}\right)^{1/2} = (M^2L^4T^{-4})^{1/2} = ML^2T^{-2} \quad (4.167)$$

4.5 The Cosmological Constant

4.5.1 Vacuum Energy and Lorentz Invariance

$$T_{\hat{\mu}\hat{\nu}}^{(vac)} = -\rho_{vac}\eta_{\hat{\mu}\hat{\nu}}, \quad (4.168)$$

$$T_{\mu\nu}^{(vac)} = -\rho_{vac}g_{\mu\nu}. \quad (4.169)$$

4.5.2 Equation of State for the Vacuum

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}. \quad (4.170)$$

$$p_{vac} = -\rho_{vac}. \quad (4.171)$$

4.5.3 Modifying Einstein's Equation

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{(M)} + T_{\mu\nu}^{(vac)}\right) \\ R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{(M)} - \rho_{vac}g_{\mu\nu}\right). \quad (4.172)$$

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + 8\pi G\rho_{vac}g_{\mu\nu} = 8\pi GT_{\mu\nu}^{(M)}. \quad (4.173)$$

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi GT_{\mu\nu}. \quad (4.174)$$

$$\rho_{vac} = \frac{\Lambda}{8\pi G}. \quad (4.175)$$

4.5.4 Lagrangian Derivation

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G} (R - 2\Lambda) + \hat{\mathcal{L}} * M \right]. \quad (4.176)$$

$$\begin{aligned} \delta S * \Lambda &= \delta \int d^4x \sqrt{-g} \left(\frac{-2\Lambda}{16\pi G} \right) \\ &= \int d^4x \left(\frac{-2\Lambda}{16\pi G} \right) \left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \right) \\ &= \frac{1}{16\pi G} \int d^4x \sqrt{-g} (\Lambda g_{\mu\nu}) \delta g^{\mu\nu}. \end{aligned} \quad (4.177)$$

4.5.5 The Cosmological Constant Problem

$$\rho_{vac} = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2} \hbar \sqrt{m^2 + k^2}. \quad (4.178)$$

$$\rho_{vac} \sim \hbar k_{max}^4. \quad (4.179)$$

$$\rho_{vac}^{(theory)} \sim (10^{18} \text{ GeV})^4 \sim 10^{112} \text{ erg/cm}^3. \quad (4.180)$$

$$|\rho_{\Lambda}^{(obs)}| \leq (10^{-12} \text{ GeV})^4 \sim 10^{-8} \text{ erg/cm}^3. \quad (4.181)$$

4.6 Energy Conditions

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}, \quad (4.182)$$

4.6.1 The Weak Energy Condition (WEC)

$$T_{\mu\nu} t^\mu t^\nu \geq 0 \quad \text{for all timelike vectors } t^\mu. \quad (4.183)$$

$$-(t^0)^2 + (t^i)^2 < 0 \implies (t^i)^2 < (t^0)^2, \quad (4.184)$$

$$\begin{aligned} T_{\mu\nu} t^\mu t^\nu &= T_{00} (t^0)^2 + T_{ij} t^i t^j \\ &= \rho (t^0)^2 + p (t^i)^2 \\ &= (t^0)^2 \left[\rho + p \frac{(t^i)^2}{(t^0)^2} \right] \\ &= (t^0)^2 (\rho + p v^2). \end{aligned} \quad (4.185)$$

$$\rho \geq 0 \quad \text{and} \quad \rho + p \geq 0. \quad (4.186)$$

4.6.2 The Null Energy Condition (NEC)

$$T_{\mu\nu} l^\mu l^\nu \geq 0 \quad \text{for all null vectors } l^\mu. \quad (4.187)$$

$$\begin{aligned} T_{\mu\nu} l^\mu l^\nu &= \rho (l^0)^2 + p (l^i)^2 \\ &= \rho (l^0)^2 + p (l^0)^2 \\ &= (l^0)^2 (\rho + p). \end{aligned} \quad (4.188)$$

$$\rho + p \geq 0. \quad (4.189)$$

4.6.3 The Dominant Energy Condition (DEC)

$$T_{\mu\nu}t^\mu t^\nu \geq 0 \quad \text{and} \quad P_\mu P^\mu \leq 0 \quad \text{for all timelike } t^\mu. \quad (4.190)$$

$$T^0 * 0 = g^{00}T * 00 = (-1)(\rho) = -\rho, \quad T^i * j = g^{ik}T * kj = \delta^{ik}(p\delta_{kj}) = p\delta^i * j. \quad (4.191)$$

$$P^0 = -T^0 * \nu t^\nu = -T^0 * 0t^0 = -(-\rho)t^0 = \rho t^0 \quad (4.192)$$

$$P^i = -T^i * \nu t^\nu = -T^i * jt^j = -pt^i. \quad (4.193)$$

$$\begin{aligned} P * \mu P^\mu &= -(P^0)^2 + (P^i)^2 \\ &= -(\rho t^0)^2 + (-pt^i)^2 \\ &= -\rho^2(t^0)^2 + p^2(t^i)^2. \end{aligned} \quad (4.194)$$

$$p^2(t^i)^2 \leq \rho^2(t^0)^2 \implies \frac{p^2}{\rho^2} \leq \frac{(t^0)^2}{(t^i)^2}. \quad (4.195)$$

$$\frac{p^2}{\rho^2} \leq 1 \implies p^2 \leq \rho^2. \quad (4.196)$$

$$\rho \geq |p|. \quad (4.197)$$

4.6.4 The Null Dominant Energy Condition (NDEC)

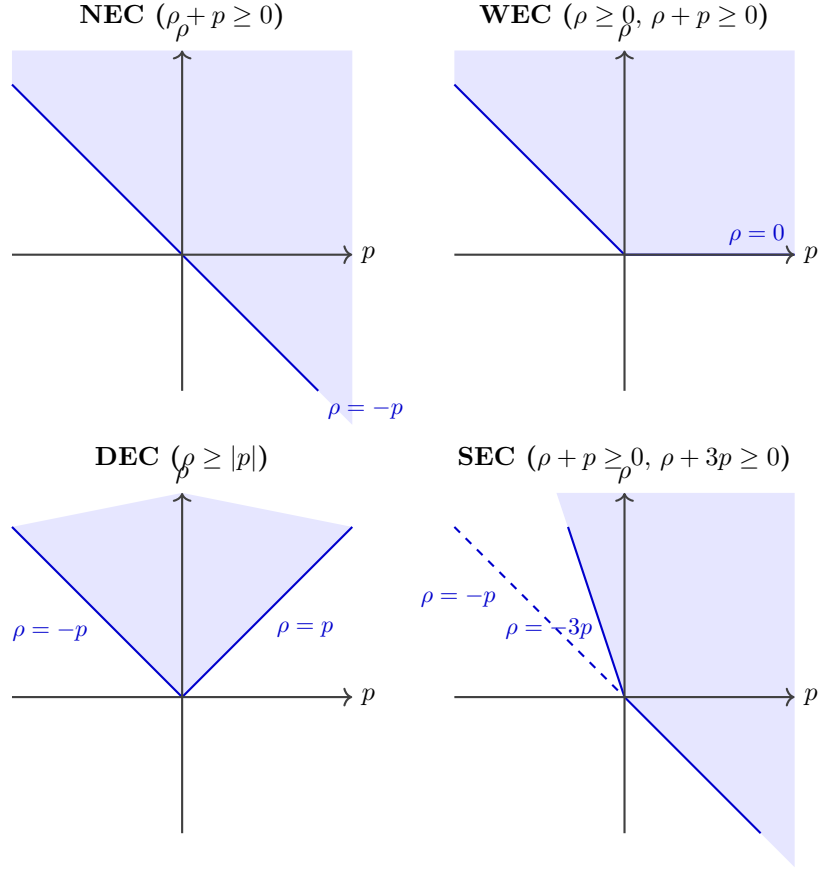
$$T_{\mu\nu}l^\mu l^\nu \geq 0 \quad \text{and} \quad P_\mu P^\mu \leq 0 \quad \text{where } P^\mu = -T^\mu * \nu l^\nu. \quad (4.198)$$

$$\begin{aligned} P * \mu P^\mu &= -\rho^2(l^0)^2 + p^2(l^i)^2 \\ &= (l^0)^2(p^2 - \rho^2). \end{aligned} \quad (4.199)$$

4.6.5 The Strong Energy Condition (SEC)

$$T_{\mu\nu}t^\mu t^\nu \geq \frac{1}{2}Tt^\sigma t_\sigma \quad \text{for all timelike vectors } t^\mu. \quad (4.200)$$

$$\begin{aligned} (t^0)^2(\rho + pv^2) &\geq \frac{1}{2}(-\rho + 3p) [-(t^0)^2(1 - v^2)] \\ \rho + pv^2 &\geq \frac{1}{2}(\rho - 3p)(1 - v^2). \end{aligned} \quad (4.201)$$



The SEC and Gravitational Attraction

$$\begin{aligned}
 R_{\mu\nu}t^\mu t^\nu &= 8\pi G \left(T_{\mu\nu}t^\mu t^\nu - \frac{1}{2}Tg_{\mu\nu}t^\mu t^\nu \right) \\
 &= 8\pi G \left(T_{\mu\nu}t^\mu t^\nu - \frac{1}{2}Tt_\sigma t^\sigma \right).
 \end{aligned} \tag{4.202}$$

$$T_{\mu\nu}t^\mu t^\nu - \frac{1}{2}Tt_\sigma t^\sigma \geq 0 \implies T_{\mu\nu}t^\mu t^\nu \geq \frac{1}{2}Tt_\sigma t^\sigma. \tag{4.203}$$

4.7 The Equivalence Principle Revisited

4.7.1 The Approximation of Minimal Coupling

$$S_{\min} = \int d^4x \sqrt{-g} \left[-\frac{1}{2}g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi - \frac{1}{2}m^2\phi^2 \right]. \tag{4.204}$$

$$S_{\text{non-min}} = \int d^4x \sqrt{-g} \left[\frac{1}{2}\xi R\phi^2 \right], \tag{4.205}$$

4.8 Alternative Theories of Gravity

4.8.1 Higher-Order Actions: $f(R)$ Gravity

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G}f(R) + \mathcal{L} * M \right]. \tag{4.206}$$

$$\delta S * f(R) = \frac{1}{16\pi G} \int d^4x \left[\delta(\sqrt{-g})f(R) + \sqrt{-g}\delta(f(R)) \right]. \tag{4.207}$$

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[-\frac{1}{2} f(R) g_{\mu\nu} \delta g^{\mu\nu} + f'(R) \delta R \right], \quad (4.208)$$

$$\delta R = R_{\mu\nu} \delta g^{\mu\nu} + g^{\mu\nu} \delta R_{\mu\nu}. \quad (4.209)$$

$$\delta R_{\mu\nu} = \nabla_\lambda (\delta \Gamma^\lambda_{\mu\nu}) - \nabla_\mu (\delta \Gamma^\lambda_{\lambda\nu}) - \nabla_\nu (\delta \Gamma^\lambda_{\lambda\mu}). \quad (4.210)$$

$$g^{\mu\nu} \delta R_{\mu\nu} = \nabla_\mu \nabla_\nu (\delta g^{\mu\nu}) - \square (g_{\mu\nu} \delta g^{\mu\nu}), \quad (4.211)$$

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} \right] \delta g^{\mu\nu} + \frac{1}{16\pi G} \int d^4x \sqrt{-g} f'(R) [\nabla_\mu \nabla_\nu - g_{\mu\nu} \square] \delta g^{\mu\nu}. \quad (4.212)$$

$$\int d^4x \sqrt{-g} f'(R) \nabla_\mu \nabla_\nu (\delta g^{\mu\nu}) = \int d^4x \sqrt{-g} [\nabla_\mu \nabla_\nu f'(R)] \delta g^{\mu\nu}, \quad (4.213)$$

$$\delta S_{f(R)} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + \nabla_\mu \nabla_\nu f'(R) - g_{\mu\nu} \square f'(R) \right] \delta g^{\mu\nu}. \quad (4.214)$$

$$f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + (\nabla_\mu \nabla_\nu - g_{\mu\nu} \square) f'(R) = 8\pi G T_{\mu\nu}. \quad (4.215)$$

4.8.2 Scalar-Tensor Theories

$$S = \int d^4x \sqrt{-g} \left[\phi R - \frac{\omega(\phi)}{\phi} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) + \mathcal{L}_M \right], \quad (4.216)$$

4.8.3 Torsion and Non-Christoffel Connections

$$T^\rho * \mu\nu = \Gamma^\rho * \mu\nu - \Gamma^\rho_{\nu\mu}. \quad (4.217)$$

4.8.4 Additional Tensor Fields and Geometric Generalizations

$$\Gamma^\rho * \mu\nu = \tilde{\Gamma}^\rho * \mu\nu + C^\rho_{\mu\nu}. \quad (4.218)$$

4.9 Variational Derivations of Advanced Field Equations

4.9.1 The Electromagnetic Stress-Energy Tensor

$$S_{\text{EM}} = \int d^4x \sqrt{-g} \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + A_\mu J^\mu \right), \quad (4.219)$$

$$F^{\mu\nu} F_{\mu\nu} = g^{\mu\alpha} g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta}. \quad (4.220)$$

$$\begin{aligned} \delta S_{\text{EM}} &= \delta \int d^4x \left[\sqrt{-g} \left(-\frac{1}{4} g^{\mu\alpha} g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta} \right) \right] \\ &= \int d^4x \left[(\delta \sqrt{-g}) \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} \right) - \frac{1}{4} \sqrt{-g} \delta (g^{\mu\alpha} g^{\nu\beta}) F_{\mu\nu} F_{\alpha\beta} \right]. \end{aligned} \quad (4.221)$$

$$\delta (g^{\mu\alpha} g^{\nu\beta}) = (\delta g^{\mu\alpha}) g^{\nu\beta} + g^{\mu\alpha} (\delta g^{\nu\beta}). \quad (4.222)$$

$$-\frac{1}{4} \sqrt{-g} \left[(\delta g^{\mu\alpha}) g^{\nu\beta} F_{\mu\nu} F_{\alpha\beta} + g^{\mu\alpha} (\delta g^{\nu\beta}) F_{\mu\nu} F_{\alpha\beta} \right]. \quad (4.223)$$

$$-\frac{1}{4} \sqrt{-g} \left[\delta g^{\rho\sigma} (g^{\nu\beta} F_{\rho\nu} F_{\sigma\beta}) + \delta g^{\rho\sigma} (g^{\mu\alpha} F_{\mu\rho} F_{\alpha\sigma}) \right]. \quad (4.224)$$

$$-\frac{1}{4} \sqrt{-g} \left[2 \delta g^{\rho\sigma} F_\rho{}^\beta F_{\sigma\beta} \right] = -\frac{1}{2} \sqrt{-g} F_{\rho\lambda} F_\sigma{}^\lambda \delta g^{\rho\sigma}. \quad (4.225)$$

$$\delta S_{\text{EM}} = \int d^4x \sqrt{-g} \left[\frac{1}{8} g_{\rho\sigma} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} F_{\rho\lambda} F_\sigma{}^\lambda \right] \delta g^{\rho\sigma}. \quad (4.226)$$

$$T_{\rho\sigma} = F_{\rho\lambda} F_\sigma{}^\lambda - \frac{1}{4} g_{\rho\sigma} F_{\mu\nu} F^{\mu\nu}. \quad (4.227)$$

4.9.2 The Palatini Formulation of General Relativity

$$S[g, \Gamma] = \int d^4x \sqrt{-g} g^{\mu\nu} R_{\mu\nu}(\Gamma). \quad (4.228)$$

$$\delta R_{\mu\nu} = \nabla_\lambda (\delta \Gamma^\lambda * \mu\nu) - \nabla * \nu (\delta \Gamma^\lambda * \mu\lambda). \quad (4.229)$$

$$\delta * \Gamma S = \int d^4x \sqrt{-g} g^{\mu\nu} \left[\nabla_\lambda (\delta \Gamma^\lambda * \mu\nu) - \nabla * \nu (\delta \Gamma^\lambda * \mu\lambda) \right] = 0. \quad (4.230)$$

$$\int d^4x \sqrt{-g} f \nabla * \lambda V^\lambda = - \int d^4x \sqrt{-g} V^\lambda \nabla_\lambda f + \text{boundary terms}. \quad (4.231)$$

$$\delta_\Gamma S = \int d^4x \left[-\nabla_\lambda (\sqrt{-g} g^{\mu\nu}) \delta \Gamma^\lambda * \mu\nu + \nabla * \nu (\sqrt{-g} g^{\mu\nu}) \delta \Gamma^\lambda * \mu\lambda \right]. \quad (4.232)$$

$$\delta * \Gamma S = \int d^4x \left[-\nabla_\lambda (\sqrt{-g} g^{\mu\nu}) + \delta^\nu * \lambda \nabla * \sigma (\sqrt{-g} g^{\mu\sigma}) \right] \delta \Gamma^\lambda * \mu\nu = 0. \quad (4.233)$$

$$\nabla * \lambda (\sqrt{-g} g^{\mu\nu}) - \delta^\nu * \lambda \nabla * \sigma (\sqrt{-g} g^{\mu\sigma}) = 0. \quad (4.234)$$

$$\nabla_\sigma (\sqrt{-g} g^{\mu\sigma}) - 4 \nabla_\sigma (\sqrt{-g} g^{\mu\sigma}) = 0 \implies -3 \nabla_\sigma (\sqrt{-g} g^{\mu\sigma}) = 0. \quad (4.235)$$

$$\nabla_\lambda (\sqrt{-g} g^{\mu\nu}) = 0. \quad (4.236)$$

$$\sqrt{-g} \nabla_\lambda g^{\mu\nu} + g^{\mu\nu} \nabla_\lambda \sqrt{-g} = 0. \quad (4.237)$$

$$\nabla_\lambda g^{\mu\nu} = 0. \quad (4.238)$$

Chapter 5

The Schwarzschild Solution

5.1 The Schwarzschild Solution

5.1.1 Metric Ansatz and Symmetry Justification

To describe the exterior gravitational field of a static, spherically symmetric mass, we start with Minkowski space:

$$ds_{\text{Minkowski}}^2 = -dt^2 + dr^2 + r^2 d\Omega^2, \quad (5.1)$$

$$d\Omega^2 = d\theta^2 + \sin^2 \theta, d\phi^2. \quad (5.2)$$

Imposing a static condition requires t -independence and time-reversal invariance ($t \rightarrow -t, dt \rightarrow -dt$) on a generic metric:

$$ds^2 = g_{tt}dt^2 + 2g_{ti}dtdx^i + g_{ij}dx^i dx^j. \quad (5.3)$$

$$ds^2 = g_{tt}(-dt)^2 + 2g_{ti}(-dt)dx^i + g_{ij}dx^i dx^j, \quad (5.4)$$

$$ds^2 = g_{tt}dt^2 - 2g_{ti}dtdx^i + g_{ij}dx^i dx^j. \quad (5.5)$$

Invariance demands:

$$2g_{ti}dtdx^i = -2g_{ti}dtdx^i. \quad (5.6)$$

Thus $g_{ti} = 0$. Spherical symmetry limits angular dependence to $d\Omega^2$. To maintain a Lorentzian signature, we write the remaining components as exponentials:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + e^{2\gamma(r)}r^2 d\Omega^2. \quad (5.7)$$

Absorbing $e^{2\gamma(r)}$ into a new radial coordinate $\bar{r} = e^{\gamma(r)}r$:

$$\bar{r} = e^{\gamma(r)}r. \quad (5.8)$$

$$d\bar{r} = \frac{d}{dr} \left(e^{\gamma(r)}r \right) dr, \quad (5.9)$$

$$d\bar{r} = \left(\frac{d}{dr} \left(e^{\gamma(r)} \right) r + e^{\gamma(r)} \frac{d}{dr} (r) \right) dr, \quad (5.10)$$

$$d\bar{r} = \left(e^{\gamma(r)} \frac{d\gamma(r)}{dr} r + e^{\gamma(r)} \right) dr. \quad (5.11)$$

$$d\bar{r} = e^{\gamma(r)} \left(1 + r \frac{d\gamma(r)}{dr} \right) dr. \quad (5.12)$$

$$dr = e^{-\gamma(r)} \left(1 + r \frac{d\gamma(r)}{dr} \right)^{-1} d\bar{r}. \quad (5.13)$$

$$dr^2 = e^{-2\gamma(r)} \left(1 + r \frac{d\gamma(r)}{dr} \right)^{-2} d\bar{r}^2. \quad (5.14)$$

Substituting back:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)} \left[e^{-2\gamma(r)} \left(1 + r \frac{d\gamma(r)}{dr} \right)^{-2} d\bar{r}^2 \right] + \bar{r}^2 d\Omega^2, \quad (5.15)$$

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)-2\gamma(r)} \left(1 + r \frac{d\gamma(r)}{dr} \right)^{-2} d\bar{r}^2 + \bar{r}^2 d\Omega^2. \quad (5.16)$$

Grouping factors:

$$e^{2\bar{\beta}(\bar{r})} = e^{2\beta(r)-2\gamma(r)} \left(1 + r \frac{d\gamma(r)}{dr}\right)^{-2}, \quad (5.17)$$

$$ds^2 = -e^{2\bar{\alpha}(\bar{r})} dt^2 + e^{2\bar{\beta}(\bar{r})} d\bar{r}^2 + \bar{r}^2 d\Omega^2. \quad (5.18)$$

Dropping overbars gives the simplified general ansatz:

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2. \quad (5.19)$$

5.1.2 Inverse Metric and Determinant

Covariant Metric Tensor Components

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2. \quad (5.20)$$

$$g_{tt} = -e^{2\alpha(r)}, \quad (5.21)$$

$$g_{rr} = e^{2\beta(r)}, \quad (5.22)$$

$$g_{\theta\theta} = r^2, \quad (5.23)$$

$$g_{\phi\phi} = r^2 \sin^2 \theta. \quad (5.24)$$

$$g_{\mu\nu} = 0 \quad \text{for} \quad \mu \neq \nu. \quad (5.25)$$

$$g_{\mu\nu} = \begin{pmatrix} -e^{2\alpha(r)} & 0 & 0 & 0 \\ 0 & e^{2\beta(r)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}. \quad (5.26)$$

Computation of the Inverse Metric Tensor

$$g^{\mu\lambda} g_{\lambda\nu} = \delta_\nu^\mu, \quad (5.27)$$

$$\sum_\lambda g^{t\lambda} g_{\lambda t} = \delta_t^t. \quad (5.28)$$

$$g^{tt} g_{tt} + g^{tr} g_{rt} + g^{t\theta} g_{\theta t} + g^{t\phi} g_{\phi t} = 1. \quad (5.29)$$

$$g^{tt} g_{tt} = 1. \quad (5.30)$$

$$g^{tt} \left(-e^{2\alpha(r)}\right) = 1. \quad (5.31)$$

$$g^{tt} = \frac{1}{-e^{2\alpha(r)}}. \quad (5.32)$$

$$g^{tt} = -e^{-2\alpha(r)}. \quad (5.33)$$

$$g^{rr} g_{rr} = 1, \quad (5.34)$$

$$g^{\theta\theta} g_{\theta\theta} = 1, \quad (5.35)$$

$$g^{\phi\phi} g_{\phi\phi} = 1. \quad (5.36)$$

$$g^{rr} = \frac{1}{g_{rr}} = \frac{1}{e^{2\beta(r)}} = e^{-2\beta(r)}, \quad (5.37)$$

$$g^{\theta\theta} = \frac{1}{g_{\theta\theta}} = \frac{1}{r^2}, \quad (5.38)$$

$$g^{\phi\phi} = \frac{1}{g_{\phi\phi}} = \frac{1}{r^2 \sin^2 \theta}. \quad (5.39)$$

$$\sum_{\lambda} g^{t\lambda} g_{\lambda r} = \delta_r^t. \quad (5.40)$$

$$g^{tt} g_{tr} + g^{tr} g_{rr} + g^{t\theta} g_{\theta r} + g^{t\phi} g_{\phi r} = 0. \quad (5.41)$$

$$g^{tr} g_{rr} = 0. \quad (5.42)$$

$$g^{tr} = 0. \quad (5.43)$$

$$g^{\mu\nu} = \begin{pmatrix} -e^{-2\alpha(r)} & 0 & 0 & 0 \\ 0 & e^{-2\beta(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2 \theta} \end{pmatrix}. \quad (5.44)$$

Verification of the Inverse Metric via Matrix Multiplication

$$g^{\mu\lambda} g_{\lambda\nu} = \begin{pmatrix} -e^{-2\alpha(r)} & 0 & 0 & 0 \\ 0 & e^{-2\beta(r)} & 0 & 0 \\ 0 & 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & 0 & \frac{1}{r^2 \sin^2 \theta} \end{pmatrix} \begin{pmatrix} -e^{2\alpha(r)} & 0 & 0 & 0 \\ 0 & e^{2\beta(r)} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}. \quad (5.45)$$

$$(g \cdot g) * tt = (-e^{-2\alpha(r)}) (-e^{2\alpha(r)}) + 0 + 0 + 0, \quad (5.46)$$

$$(g \cdot g) * rr = 0 + (e^{-2\beta(r)}) (e^{2\beta(r)}) + 0 + 0, \quad (5.47)$$

$$(g \cdot g) * \theta\theta = 0 + 0 + \left(\frac{1}{r^2}\right) (r^2) + 0, \quad (5.48)$$

$$(g \cdot g) * \phi\phi = 0 + 0 + 0 + \left(\frac{1}{r^2 \sin^2 \theta}\right) (r^2 \sin^2 \theta). \quad (5.49)$$

$$(g \cdot g) * tt = e^{-2\alpha(r)+2\alpha(r)} = e^0 = 1, \quad (5.50)$$

$$(g \cdot g) * rr = e^{-2\beta(r)+2\beta(r)} = e^0 = 1, \quad (5.51)$$

$$(g \cdot g) * \theta\theta = 1, \quad (5.52)$$

$$(g \cdot g) * \phi\phi = 1. \quad (5.53)$$

$$g^{\mu\lambda} g_{\lambda\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \delta_{\nu}^{\mu}. \quad (5.54)$$

Computation of the Metric Determinant

$$g = g_{tt} g_{rr} g_{\theta\theta} g_{\phi\phi}. \quad (5.55)$$

$$g = (-e^{2\alpha(r)}) (e^{2\beta(r)}) (r^2) (r^2 \sin^2 \theta). \quad (5.56)$$

$$g = -(e^{2\alpha(r)} e^{2\beta(r)}) (r^2 \cdot r^2) (\sin^2 \theta). \quad (5.57)$$

$$g = -e^{2\alpha(r)+2\beta(r)} r^4 \sin^2 \theta. \quad (5.58)$$

$$g = -e^{2(\alpha(r)+\beta(r))} r^4 \sin^2 \theta. \quad (5.59)$$

Computation of the Invariant Volume Element Factor

$$-g = - \left(-e^{2(\alpha(r)+\beta(r))} r^4 \sin^2 \theta \right). \quad (5.60)$$

$$-g = e^{2(\alpha(r)+\beta(r))} r^4 \sin^2 \theta. \quad (5.61)$$

$$\sqrt{-g} = \sqrt{e^{2(\alpha(r)+\beta(r))} r^4 \sin^2 \theta}. \quad (5.62)$$

$$\sqrt{-g} = \left(e^{2(\alpha(r)+\beta(r))} \right)^{\frac{1}{2}} (r^4)^{\frac{1}{2}} (\sin^2 \theta)^{\frac{1}{2}}. \quad (5.63)$$

$$\left(e^{2(\alpha(r)+\beta(r))} \right)^{\frac{1}{2}} = e^{\alpha(r)+\beta(r)}, \quad (5.64)$$

$$(r^4)^{\frac{1}{2}} = r^2, \quad (5.65)$$

$$(\sin^2 \theta)^{\frac{1}{2}} = \sin \theta. \quad (5.66)$$

$$\sqrt{-g} = e^{\alpha(r)+\beta(r)} r^2 \sin \theta. \quad (5.67)$$

Deriving Christoffel Symbols via the Lagrangian Method

Using $\mathcal{L} = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$:

$$\mathcal{L} = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \quad (5.68)$$

$$\mathcal{L} = -e^{2\alpha(t,r)} \dot{t}^2 + e^{2\beta(t,r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2 \quad (5.69)$$

Euler-Lagrange equations yield geodesics:

$$\frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) - \frac{\partial \mathcal{L}}{\partial x^\mu} = 0 \quad (5.70)$$

$$\ddot{x}^\mu + \Gamma_{\rho\sigma}^\mu \dot{x}^\rho \dot{x}^\sigma = 0 \quad (5.71)$$

1. The t-equation

$$\frac{\partial \mathcal{L}}{\partial \dot{t}} = -2e^{2\alpha(t,r)} \dot{t} \quad (5.72)$$

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{t}} \right) &= -2 \frac{d}{d\lambda} \left(e^{2\alpha(t,r)} \right) \dot{t} - 2e^{2\alpha(t,r)} \ddot{t} \\ &= -2 \left(e^{2\alpha(t,r)} \cdot 2 \left[\frac{\partial \alpha}{\partial t} \dot{t} + \frac{\partial \alpha}{\partial r} \dot{r} \right] \right) \dot{t} - 2e^{2\alpha(t,r)} \ddot{t} \\ &= -4e^{2\alpha(t,r)} \partial_t \alpha \dot{t}^2 - 4e^{2\alpha(t,r)} \partial_r \alpha \dot{t} \dot{r} - 2e^{2\alpha(t,r)} \ddot{t} \end{aligned} \quad (5.73)$$

$$\frac{\partial \mathcal{L}}{\partial t} = -2\dot{t}^2 e^{2\alpha(t,r)} \partial_t \alpha + 2\dot{r}^2 e^{2\beta(t,r)} \partial_t \beta \quad (5.74)$$

$$-2e^{2\alpha(t,r)} \ddot{t} - 4e^{2\alpha(t,r)} \partial_t \alpha \dot{t}^2 - 4e^{2\alpha(t,r)} \partial_r \alpha \dot{t} \dot{r} - \left(-2e^{2\alpha(t,r)} \partial_t \alpha \dot{t}^2 + 2e^{2\beta(t,r)} \partial_t \beta \dot{r}^2 \right) = 0 \quad (5.75)$$

$$\ddot{t} + 2\partial_t \alpha \dot{t}^2 + 2\partial_r \alpha \dot{t} \dot{r} - \partial_t \alpha \dot{t}^2 + e^{2(\beta-\alpha)} \partial_t \beta \dot{r}^2 = 0 \quad (5.76)$$

$$\ddot{t} + (\partial_t \alpha) \dot{t}^2 + e^{2(\beta-\alpha)} (\partial_t \beta) \dot{r}^2 + 2(\partial_r \alpha) \dot{t} \dot{r} = 0 \quad (5.77)$$

$$\ddot{t} + \Gamma_{tt}^t \dot{t}^2 + \Gamma_{rr}^t \dot{r}^2 + \Gamma_{\theta\theta}^t \dot{\theta}^2 + \Gamma_{\phi\phi}^t \dot{\phi}^2 + 2\Gamma_{tr}^t \dot{t} \dot{r} + \dots = 0 \quad (5.78)$$

$$\Gamma_{tt}^t = \partial_t \alpha \quad (5.79)$$

$$\Gamma_{rr}^t = e^{2(\beta-\alpha)} \partial_t \beta \quad (5.80)$$

$$\Gamma_{tr}^t = \partial_r \alpha \quad (5.81)$$

2. The r -equation

$$\frac{\partial \mathcal{L}}{\partial \dot{r}} = 2e^{2\beta(t,r)} \dot{r} \quad (5.82)$$

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) &= 2 \left(e^{2\beta(t,r)} \cdot 2 \left[\frac{\partial \beta}{\partial t} \dot{t} + \frac{\partial \beta}{\partial r} \dot{r} \right] \right) \dot{r} + 2e^{2\beta(t,r)} \ddot{r} \\ &= 4e^{2\beta(t,r)} \partial_t \beta, \dot{t} \dot{r} + 4e^{2\beta(t,r)} \partial_r \beta, \dot{r}^2 + 2e^{2\beta(t,r)} \ddot{r} \end{aligned} \quad (5.83)$$

$$\frac{\partial \mathcal{L}}{\partial r} = -2\dot{t}^2 e^{2\alpha(t,r)} \partial_r \alpha + 2\dot{r}^2 e^{2\beta(t,r)} \partial_r \beta + 2r\dot{\theta}^2 + 2r \sin^2 \theta \dot{\phi}^2 \quad (5.84)$$

$$2e^{2\beta} \ddot{r} + 4e^{2\beta} \partial_t \beta, \dot{t} \dot{r} + 4e^{2\beta} \partial_r \beta, \dot{r}^2 - \left(-2e^{2\alpha} \partial_r \alpha, \dot{t}^2 + 2e^{2\beta} \partial_r \beta, \dot{r}^2 + 2r\dot{\theta}^2 + 2r \sin^2 \theta \dot{\phi}^2 \right) = 0 \quad (5.85)$$

$$\ddot{r} + 2\partial_t \beta, \dot{t} \dot{r} + 2\partial_r \beta, \dot{r}^2 + e^{2(\alpha-\beta)} \partial_r \alpha, \dot{t}^2 - \partial_r \beta, \dot{r}^2 - r e^{-2\beta} \dot{\theta}^2 - r \sin^2 \theta e^{-2\beta} \dot{\phi}^2 = 0 \quad (5.86)$$

$$\ddot{r} + e^{2(\alpha-\beta)} (\partial_r \alpha) \dot{t}^2 + (\partial_r \beta) \dot{r}^2 - r e^{-2\beta} \dot{\theta}^2 - r \sin^2 \theta e^{-2\beta} \dot{\phi}^2 + 2(\partial_t \beta) \dot{t} \dot{r} = 0 \quad (5.87)$$

$$\ddot{r} + \Gamma_{tt}^r \dot{t}^2 + \Gamma_{rr}^r \dot{r}^2 + \Gamma_{\theta\theta}^r \dot{\theta}^2 + \Gamma_{\phi\phi}^r \dot{\phi}^2 + 2\Gamma_{tr}^r \dot{t} \dot{r} + \dots = 0 \quad (5.88)$$

$$\Gamma_{tt}^r = e^{2(\alpha-\beta)} \partial_r \alpha \quad (5.89)$$

$$\Gamma_{rr}^r = \partial_r \beta \quad (5.90)$$

$$\Gamma_{\theta\theta}^r = -r e^{-2\beta} \quad (5.91)$$

$$\Gamma_{\phi\phi}^r = -r e^{-2\beta} \sin^2 \theta \quad (5.92)$$

$$\Gamma_{tr}^r = \partial_t \beta \quad (5.93)$$

3. The θ -equation

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = 2r^2 \dot{\theta} \quad (5.94)$$

$$\frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = 4r\dot{r}\dot{\theta} + 2r^2 \ddot{\theta} \quad (5.95)$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = 2r^2 \sin \theta \cos \theta, \dot{\phi}^2 \quad (5.96)$$

$$2r^2 \ddot{\theta} + 4r\dot{r}\dot{\theta} - 2r^2 \sin \theta \cos \theta, \dot{\phi}^2 = 0 \quad (5.97)$$

$$\ddot{\theta} + (-\sin \theta \cos \theta) \dot{\phi}^2 + 2 \left(\frac{1}{r} \right) \dot{r} \dot{\theta} = 0 \quad (5.98)$$

$$\ddot{\theta} + \Gamma_{\phi\phi}^\theta \dot{\phi}^2 + 2\Gamma_{r\theta}^\theta \dot{r} \dot{\theta} + \dots = 0 \quad (5.99)$$

$$\Gamma_{\phi\phi}^\theta = -\sin \theta \cos \theta \quad (5.100)$$

$$\Gamma_{r\theta}^\theta = \frac{1}{r} \quad (5.101)$$

4. The ϕ -equation

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 2r^2 \sin^2 \theta, \dot{\phi} \quad (5.102)$$

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) &= 2 \frac{d}{d\lambda} (r^2 \sin^2 \theta) \dot{\phi} + 2r^2 \sin^2 \theta, \ddot{\phi} \\ &= 2 \left(2r\dot{r} \sin^2 \theta + r^2 (2 \sin \theta \cos \theta) \dot{\theta} \right) \dot{\phi} + 2r^2 \sin^2 \theta, \ddot{\phi} \\ &= 4r \sin^2 \theta, \dot{r} \dot{\phi} + 4r^2 \sin \theta \cos \theta, \dot{\theta} \dot{\phi} + 2r^2 \sin^2 \theta, \ddot{\phi} \end{aligned} \quad (5.103)$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = 0 \quad (5.104)$$

$$2r^2 \sin^2 \theta, \ddot{\phi} + 4r \sin^2 \theta, \dot{r} \dot{\phi} + 4r^2 \sin \theta \cos \theta, \dot{\theta} \dot{\phi} = 0 \quad (5.105)$$

$$\ddot{\phi} + \frac{2}{r} \dot{r} \dot{\phi} + 2 \frac{\cos \theta}{\sin \theta} \dot{\theta} \dot{\phi} = 0 \quad (5.106)$$

$$\ddot{\phi} + 2 \left(\frac{1}{r} \right) \dot{r} \dot{\phi} + 2(\cot \theta) \dot{\theta} \dot{\phi} = 0 \quad (5.107)$$

$$\ddot{\phi} + 2\Gamma_{r\phi}^\phi \dot{r} \dot{\phi} + 2\Gamma_{\theta\phi}^\phi \dot{\theta} \dot{\phi} + \dots = 0 \quad (5.108)$$

$$\Gamma_{r\phi}^\phi = \frac{1}{r} \quad (5.109)$$

$$\Gamma_{\theta\phi}^\phi = \cot \theta \quad (5.110)$$

5. Summary

$$\Gamma_{tt}^t = \partial_t \alpha \quad (5.111)$$

$$\Gamma_{rr}^t = e^{2(\beta-\alpha)} \partial_t \beta \quad (5.112)$$

$$\Gamma_{tr}^t = \partial_r \alpha \quad (5.113)$$

$$\Gamma_{tt}^r = e^{2(\alpha-\beta)} \partial_r \alpha \quad (5.114)$$

$$\Gamma_{rr}^r = \partial_r \beta \quad (5.115)$$

$$\Gamma_{tr}^r = \partial_t \beta \quad (5.116)$$

$$\Gamma_{\theta\theta}^r = -r e^{-2\beta} \quad (5.117)$$

$$\Gamma_{\phi\phi}^r = -r e^{-2\beta} \sin^2 \theta \quad (5.118)$$

$$\Gamma_{r\theta}^\theta = \frac{1}{r} \quad (5.119)$$

$$\Gamma_{\phi\phi}^\theta = -\sin \theta \cos \theta \quad (5.120)$$

$$\Gamma_{r\phi}^\phi = \frac{1}{r} \quad (5.121)$$

$$\Gamma_{\theta\phi}^\phi = \cot \theta \quad (5.122)$$

5.1.3 Derivation of the Ricci Tensor

$$R_{\mu\nu} = R^\lambda * \mu \lambda \nu = \partial * \lambda \Gamma_{\mu\nu}^\lambda - \partial_\nu \Gamma_{\mu\lambda}^\lambda + \Gamma_{\sigma\lambda}^\lambda \Gamma_{\mu\nu}^\sigma - \Gamma_{\sigma\nu}^\lambda \Gamma_{\mu\lambda}^\sigma. \quad (5.123)$$

$$\Gamma_{\mu\lambda}^\lambda = \partial_\mu \ln \sqrt{-g}. \quad (5.124)$$

$$\sqrt{-g} = e^{\alpha(r)+\beta(r)} r^2 \sin \theta. \quad (5.125)$$

$$\ln \sqrt{-g} = \alpha(r) + \beta(r) + 2 \ln r + \ln(\sin \theta). \quad (5.126)$$

$$\Gamma_{r\lambda}^\lambda = \partial_r [\alpha(r) + \beta(r) + 2 \ln r + \ln(\sin \theta)]. \quad (5.127)$$

$$\Gamma_{r\lambda}^\lambda = \partial_r \alpha + \partial_r \beta + \frac{2}{r}. \quad (5.128)$$

$$\Gamma_{\theta\lambda}^\lambda = \partial_\theta [\alpha(r) + \beta(r) + 2 \ln r + \ln(\sin \theta)]. \quad (5.129)$$

$$\Gamma_{\theta\lambda}^\lambda = \frac{1}{\sin \theta} \cos \theta = \cot \theta. \quad (5.130)$$

Temporal Component: R_{tt}

$$R_{tt} = \partial_\lambda \Gamma_{tt}^\lambda - \partial_t \Gamma_{t\lambda}^\lambda + \Gamma_{\sigma\lambda}^\lambda \Gamma_{tt}^\sigma - \Gamma_{\sigma t}^\lambda \Gamma_{t\lambda}^\sigma. \quad (5.131)$$

$$\partial_\lambda \Gamma_{tt}^\lambda = \partial_r \Gamma_{tt}^r. \quad (5.132)$$

$$\partial_r \Gamma_{tt}^r = \partial_r \left(e^{2(\alpha-\beta)} \partial_r \alpha \right). \quad (5.133)$$

$$\partial_r \Gamma_{tt}^r = 2(\partial_r \alpha - \partial_r \beta) e^{2(\alpha-\beta)} \partial_r \alpha + e^{2(\alpha-\beta)} \partial_r^2 \alpha. \quad (5.134)$$

$$\partial_r \Gamma_{tt}^r = e^{2(\alpha-\beta)} \left[2(\partial_r \alpha)^2 - 2\partial_r \alpha \partial_r \beta + \partial_r^2 \alpha \right]. \quad (5.135)$$

$$\Gamma_{\sigma\lambda}^\lambda \Gamma_{tt}^\sigma = \Gamma_{r\lambda}^\lambda \Gamma_{tt}^r. \quad (5.136)$$

$$\Gamma_{r\lambda}^\lambda \Gamma_{tt}^r = \left(\partial_r \alpha + \partial_r \beta + \frac{2}{r} \right) \left(e^{2(\alpha-\beta)} \partial_r \alpha \right). \quad (5.137)$$

$$\Gamma_{r\lambda}^\lambda \Gamma_{tt}^r = e^{2(\alpha-\beta)} \left[(\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \alpha \right]. \quad (5.138)$$

$$-\Gamma_{\sigma t}^\lambda \Gamma_{t\lambda}^\sigma = -(\Gamma_{rt}^t \Gamma_{tt}^r + \Gamma_{tt}^r \Gamma_{tr}^t). \quad (5.139)$$

$$-\Gamma_{\sigma t}^\lambda \Gamma_{t\lambda}^\sigma = -2\Gamma_{tr}^t \Gamma_{tt}^r. \quad (5.140)$$

$$-\Gamma_{\sigma t}^\lambda \Gamma_{t\lambda}^\sigma = -2(\partial_r \alpha) \left(e^{2(\alpha-\beta)} \partial_r \alpha \right) = e^{2(\alpha-\beta)} \left[-2(\partial_r \alpha)^2 \right]. \quad (5.141)$$

$$\begin{aligned} R_{tt} &= e^{2(\alpha-\beta)} \left[2(\partial_r \alpha)^2 - 2\partial_r \alpha \partial_r \beta + \partial_r^2 \alpha \right] \\ &\quad + e^{2(\alpha-\beta)} \left[(\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \alpha \right] \\ &\quad + e^{2(\alpha-\beta)} \left[-2(\partial_r \alpha)^2 \right]. \end{aligned} \quad (5.142)$$

$$R_{tt} = e^{2(\alpha-\beta)} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \alpha \right]. \quad (5.143)$$

Radial Component: R_{rr}

$$R_{rr} = \partial_\lambda \Gamma_{rr}^\lambda - \partial_r \Gamma_{r\lambda}^\lambda + \Gamma_{\sigma\lambda}^\lambda \Gamma_{rr}^\sigma - \Gamma_{\sigma r}^\lambda \Gamma_{r\lambda}^\sigma. \quad (5.144)$$

$$\partial_\lambda \Gamma_{rr}^\lambda = \partial_r \Gamma_{rr}^r = \partial_r (\partial_r \beta) = \partial_r^2 \beta. \quad (5.145)$$

$$-\partial_r \Gamma_{r\lambda}^\lambda = -\partial_r \left(\partial_r \alpha + \partial_r \beta + \frac{2}{r} \right) = -\partial_r^2 \alpha - \partial_r^2 \beta + \frac{2}{r^2}. \quad (5.146)$$

$$\Gamma_{\sigma\lambda}^\lambda \Gamma_{rr}^\sigma = \Gamma_{r\lambda}^\lambda \Gamma_{rr}^r. \quad (5.147)$$

$$\Gamma_{r\lambda}^\lambda \Gamma_{rr}^r = \left(\partial_r \alpha + \partial_r \beta + \frac{2}{r} \right) (\partial_r \beta) = \partial_r \alpha \partial_r \beta + (\partial_r \beta)^2 + \frac{2}{r} \partial_r \beta. \quad (5.148)$$

$$-\Gamma_{\sigma r}^\lambda \Gamma_{r\lambda}^\sigma = -\left(\Gamma_{tr}^t \Gamma_{rt}^t + \Gamma_{rr}^r \Gamma_{rr}^r + \Gamma_{\theta r}^\theta \Gamma_{r\theta}^\theta + \Gamma_{\phi r}^\phi \Gamma_{r\phi}^\phi \right). \quad (5.149)$$

$$-\Gamma_{\sigma r}^\lambda \Gamma_{r\lambda}^\sigma = -\left[(\partial_r \alpha)^2 + (\partial_r \beta)^2 + \left(\frac{1}{r} \right)^2 + \left(\frac{1}{r} \right)^2 \right]. \quad (5.150)$$

$$-\Gamma_{\sigma r}^{\lambda} \Gamma_{r\lambda}^{\sigma} = -(\partial_r \alpha)^2 - (\partial_r \beta)^2 - \frac{2}{r^2}. \quad (5.151)$$

$$\begin{aligned} R_{rr} &= \partial_r^2 \beta - \partial_r^2 \alpha - \partial_r^2 \beta + \frac{2}{r^2} \\ &\quad + \partial_r \alpha \partial_r \beta + (\partial_r \beta)^2 + \frac{2}{r} \partial_r \beta \\ &\quad - (\partial_r \alpha)^2 - (\partial_r \beta)^2 - \frac{2}{r^2}. \end{aligned} \quad (5.152)$$

$$R_{rr} = -\partial_r^2 \alpha - (\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \beta. \quad (5.153)$$

Polar Component: $R_{\theta\theta}$

$$R_{\theta\theta} = \partial_{\lambda} \Gamma_{\theta\theta}^{\lambda} - \partial_{\theta} \Gamma_{\theta\lambda}^{\lambda} + \Gamma_{\sigma\lambda}^{\lambda} \Gamma_{\theta\theta}^{\sigma} - \Gamma_{\sigma\theta}^{\lambda} \Gamma_{\theta\lambda}^{\sigma}. \quad (5.154)$$

$$\partial_{\lambda} \Gamma_{\theta\theta}^{\lambda} = \partial_r \Gamma_{\theta\theta}^r = \partial_r (-re^{-2\beta}). \quad (5.155)$$

$$\partial_r \Gamma_{\theta\theta}^r = -e^{-2\beta} - r(-2\partial_r \beta e^{-2\beta}) = e^{-2\beta} (2r\partial_r \beta - 1). \quad (5.156)$$

$$-\partial_{\theta} \Gamma_{\theta\lambda}^{\lambda} = -\partial_{\theta} (\cot \theta) = \csc^2 \theta. \quad (5.157)$$

$$\Gamma_{\sigma\lambda}^{\lambda} \Gamma_{\theta\theta}^{\sigma} = \Gamma_{r\lambda}^{\lambda} \Gamma_{\theta\theta}^r. \quad (5.158)$$

$$\Gamma_{r\lambda}^{\lambda} \Gamma_{\theta\theta}^r = \left(\partial_r \alpha + \partial_r \beta + \frac{2}{r} \right) (-re^{-2\beta}) = e^{-2\beta} (-r\partial_r \alpha - r\partial_r \beta - 2). \quad (5.159)$$

$$-\Gamma_{\sigma\theta}^{\lambda} \Gamma_{\theta\lambda}^{\sigma} = -\left(\Gamma_{\phi\theta}^{\phi} \Gamma_{\theta\phi}^{\phi} + \Gamma_{\theta\theta}^r \Gamma_{\theta r}^{\theta} + \Gamma_{r\theta}^{\theta} \Gamma_{\theta\theta}^r \right). \quad (5.160)$$

$$-\Gamma_{\sigma\theta}^{\lambda} \Gamma_{\theta\lambda}^{\sigma} = -\left[(\cot \theta)^2 + (-re^{-2\beta}) \left(\frac{1}{r} \right) + \left(\frac{1}{r} \right) (-re^{-2\beta}) \right]. \quad (5.161)$$

$$-\Gamma_{\sigma\theta}^{\lambda} \Gamma_{\theta\lambda}^{\sigma} = -\cot^2 \theta + 2e^{-2\beta}. \quad (5.162)$$

$$R_{\theta\theta} = e^{-2\beta} (2r\partial_r \beta - 1) + \csc^2 \theta + e^{-2\beta} (-r\partial_r \alpha - r\partial_r \beta - 2) - \cot^2 \theta + 2e^{-2\beta}. \quad (5.163)$$

$$R_{\theta\theta} = e^{-2\beta} [r\partial_r \beta - r\partial_r \alpha - 1] + 1. \quad (5.164)$$

Azimuthal Component: $R_{\phi\phi}$

$$R_{\phi\phi} = \partial_{\lambda} \Gamma_{\phi\phi}^{\lambda} - \partial_{\phi} \Gamma_{\phi\lambda}^{\lambda} + \Gamma_{\sigma\lambda}^{\lambda} \Gamma_{\phi\phi}^{\sigma} - \Gamma_{\sigma\phi}^{\lambda} \Gamma_{\phi\lambda}^{\sigma}. \quad (5.165)$$

$$\partial_{\lambda} \Gamma_{\phi\phi}^{\lambda} = \partial_r \Gamma_{\phi\phi}^r + \partial_{\theta} \Gamma_{\phi\phi}^{\theta}. \quad (5.166)$$

$$\partial_r (-re^{-2\beta} \sin^2 \theta) = \sin^2 \theta (-e^{-2\beta} + 2r\partial_r \beta e^{-2\beta}). \quad (5.167)$$

$$\partial_{\theta} (-\sin \theta \cos \theta) = -\cos^2 \theta + \sin^2 \theta. \quad (5.168)$$

$$-\partial_{\phi} \Gamma_{\phi\lambda}^{\lambda} = 0. \quad (5.169)$$

$$\Gamma_{\sigma\lambda}^{\lambda} \Gamma_{\phi\phi}^{\sigma} = \Gamma_{r\lambda}^{\lambda} \Gamma_{\phi\phi}^r + \Gamma_{\theta\lambda}^{\theta} \Gamma_{\phi\phi}^{\theta}. \quad (5.170)$$

$$\Gamma_{\sigma\lambda}^{\lambda} \Gamma_{\phi\phi}^{\sigma} = \left(\partial_r \alpha + \partial_r \beta + \frac{2}{r} \right) (-re^{-2\beta} \sin^2 \theta) + (\cot \theta) (-\sin \theta \cos \theta). \quad (5.171)$$

$$\Gamma_{\sigma\lambda}^{\lambda} \Gamma_{\phi\phi}^{\sigma} = \sin^2 \theta, e^{-2\beta} (-r\partial_r \alpha - r\partial_r \beta - 2) - \cos^2 \theta. \quad (5.172)$$

$$-\Gamma_{\sigma\phi}^{\lambda} \Gamma_{\phi\lambda}^{\sigma} = -\left(\Gamma_{\phi\phi}^r \Gamma_{\phi r}^{\phi} + \Gamma_{r\phi}^{\phi} \Gamma_{\phi\phi}^r + \Gamma_{\phi\phi}^{\theta} \Gamma_{\phi\theta}^{\phi} + \Gamma_{\theta\phi}^{\theta} \Gamma_{\phi\phi}^{\theta} \right). \quad (5.173)$$

$$-\Gamma_{\sigma\phi}^{\lambda} \Gamma_{\phi\lambda}^{\sigma} = -2\Gamma_{\phi\phi}^r \Gamma_{\phi r}^{\phi} - 2\Gamma_{\phi\phi}^{\theta} \Gamma_{\phi\theta}^{\phi}. \quad (5.174)$$

$$-\Gamma_{\sigma\phi}^{\lambda}\Gamma_{\phi\lambda}^{\sigma} = -2 \left(-re^{-2\beta} \sin^2 \theta \right) \left(\frac{1}{r} \right) - 2 (-\sin \theta \cos \theta) \left(\frac{\cos \theta}{\sin \theta} \right). \quad (5.175)$$

$$-\Gamma_{\sigma\phi}^{\lambda}\Gamma_{\phi\lambda}^{\sigma} = 2e^{-2\beta} \sin^2 \theta + 2 \cos^2 \theta. \quad (5.176)$$

$$\begin{aligned} R_{\phi\phi} &= \sin^2 \theta \left(-e^{-2\beta} + 2r\partial_r\beta e^{-2\beta} \right) - \cos^2 \theta + \sin^2 \theta \\ &\quad + \sin^2 \theta, e^{-2\beta} (-r\partial_r\alpha - r\partial_r\beta - 2) - \cos^2 \theta \\ &\quad + 2e^{-2\beta} \sin^2 \theta + 2 \cos^2 \theta. \end{aligned} \quad (5.177)$$

$$R_{\phi\phi} = \sin^2 \theta \left[e^{-2\beta} (-1 + 2r\partial_r\beta - r\partial_r\alpha - r\partial_r\beta - 2 + 2) + 1 \right]. \quad (5.178)$$

$$R_{\phi\phi} = \sin^2 \theta \left[e^{-2\beta} (r\partial_r\beta - r\partial_r\alpha - 1) + 1 \right]. \quad (5.179)$$

5.1.4 Einstein Equations and Solution

$$R_{\mu\nu} = 0. \quad (5.180)$$

$$R_{tt} = e^{2(\alpha-\beta)} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \alpha \right] = 0, \quad (5.181)$$

$$R_{rr} = -\partial_r^2 \alpha - (\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \beta = 0, \quad (5.182)$$

$$R_{\theta\theta} = e^{-2\beta} [r(\partial_r \beta - \partial_r \alpha) - 1] + 1 = 0. \quad (5.183)$$

$$e^{2(\beta-\alpha)} R_{tt} + R_{rr} = 0. \quad (5.184)$$

$$\begin{aligned} e^{2(\beta-\alpha)} R_{tt} + R_{rr} &= \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \alpha \right] \\ &\quad + \left[-\partial_r^2 \alpha - (\partial_r \alpha)^2 + \partial_r \alpha \partial_r \beta + \frac{2}{r} \partial_r \beta \right]. \end{aligned} \quad (5.185)$$

$$e^{2(\beta-\alpha)} R_{tt} + R_{rr} = \frac{2}{r} \partial_r \alpha + \frac{2}{r} \partial_r \beta = 0. \quad (5.186)$$

$$\frac{2}{r} (\partial_r \alpha + \partial_r \beta) = 0. \quad (5.187)$$

$$\partial_r (\alpha + \beta) = 0. \quad (5.188)$$

$$\alpha(r) + \beta(r) = c, \quad (5.189)$$

Absorbing the constant c :

$$\alpha(r) = -\beta(r). \quad (5.190)$$

Substituting $\beta = -\alpha$:

$$e^{-2\beta} [r(\partial_r \beta - \partial_r \alpha) - 1] + 1 = 0. \quad (5.191)$$

$$e^{2\alpha} [r(-\partial_r \alpha - \partial_r \alpha) - 1] + 1 = 0. \quad (5.192)$$

$$e^{2\alpha} (-2r\partial_r \alpha - 1) + 1 = 0. \quad (5.193)$$

$$e^{2\alpha} (2r\partial_r \alpha + 1) = 1. \quad (5.194)$$

$$\partial_r (re^{2\alpha}) = (\partial_r r) e^{2\alpha} + r (\partial_r e^{2\alpha}). \quad (5.195)$$

$$\partial_r (re^{2\alpha}) = e^{2\alpha} + r (2\partial_r \alpha, e^{2\alpha}). \quad (5.196)$$

$$\partial_r (re^{2\alpha}) = e^{2\alpha} (1 + 2r\partial_r \alpha). \quad (5.197)$$

$$\partial_r (re^{2\alpha}) = 1. \quad (5.198)$$

$$\int \partial_r (re^{2\alpha}) dr = \int 1, dr. \quad (5.199)$$

$$re^{2\alpha(r)} = r + C_1, \quad (5.200)$$

$$e^{2\alpha(r)} = 1 + \frac{C_1}{r}. \quad (5.201)$$

Setting $C_1 = -R_S$:

$$e^{2\alpha(r)} = 1 - \frac{R_S}{r}. \quad (5.202)$$

$$e^{2\beta(r)} = e^{-2\alpha(r)} = \left(1 - \frac{R_S}{r}\right)^{-1}. \quad (5.203)$$

5.1.5 Final Schwarzschild Metric and Interpretation

$$ds^2 = -\left(1 - \frac{R_S}{r}\right) dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (5.204)$$

Matching the Newtonian limit:

$$g_{tt} \approx -(1 + 2\Phi). \quad (5.205)$$

$$\Phi = -\frac{GM}{r}. \quad (5.206)$$

$$g_{tt} \approx -\left(1 - \frac{2GM}{r}\right). \quad (5.207)$$

$$g_{tt} = -\left(1 - \frac{R_S}{r}\right). \quad (5.208)$$

$$-\left(1 - \frac{R_S}{r}\right) = -\left(1 - \frac{2GM}{r}\right). \quad (5.209)$$

$$-\frac{R_S}{r} = -\frac{2GM}{r}. \quad (5.210)$$

$$R_S = 2GM. \quad (5.211)$$

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (5.212)$$

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta, d\phi^2. \quad (5.213)$$

5.2 Birkhoff's Theorem

5.2.1 General Spherically Symmetric Metric

$$ds^2 = g_{aa}(a, b)da^2 + g_{ab}(a, b)(dad b + dbda) + g_{bb}(a, b)db^2 + r^2(a, b)d\Omega^2 \quad (5.214)$$

$$ds^2 = g_{aa}(a, b)da^2 + 2g_{ab}(a, b)dad b + g_{bb}(a, b)db^2 + r^2(a, b)d\Omega^2 \quad (5.215)$$

Changing coordinates to (a, r) :

$$db = \frac{\partial b}{\partial a} da + \frac{\partial b}{\partial r} dr \quad (5.216)$$

$$\begin{aligned} db^2 &= \left(\frac{\partial b}{\partial a} da + \frac{\partial b}{\partial r} dr\right)^2 \\ &= \left(\frac{\partial b}{\partial a}\right)^2 da^2 + 2\left(\frac{\partial b}{\partial a}\right)\left(\frac{\partial b}{\partial r}\right) dad r + \left(\frac{\partial b}{\partial r}\right)^2 dr^2 \end{aligned} \quad (5.217)$$

$$\begin{aligned}
2g_{ab}dad b &= 2g_{ab}da \left(\frac{\partial b}{\partial a}da + \frac{\partial b}{\partial r}dr \right) \\
&= 2g_{ab}\frac{\partial b}{\partial a}da^2 + 2g_{ab}\frac{\partial b}{\partial r}dad r
\end{aligned} \tag{5.218}$$

$$\begin{aligned}
ds^2 &= g_{aa}da^2 + \left[2g_{ab}\frac{\partial b}{\partial a}da^2 + 2g_{ab}\frac{\partial b}{\partial r}dad r \right] \\
&+ g_{bb} \left[\left(\frac{\partial b}{\partial a} \right)^2 da^2 + 2 \left(\frac{\partial b}{\partial a} \right) \left(\frac{\partial b}{\partial r} \right) dad r + \left(\frac{\partial b}{\partial r} \right)^2 dr^2 \right] + r^2 d\Omega^2
\end{aligned} \tag{5.219}$$

$$\begin{aligned}
ds^2 &= \left[g_{aa} + 2g_{ab}\frac{\partial b}{\partial a} + g_{bb} \left(\frac{\partial b}{\partial a} \right)^2 \right] da^2 \\
&+ 2 \left[g_{ab}\frac{\partial b}{\partial r} + g_{bb} \left(\frac{\partial b}{\partial a} \right) \left(\frac{\partial b}{\partial r} \right) \right] dad r \\
&+ \left[g_{bb} \left(\frac{\partial b}{\partial r} \right)^2 \right] dr^2 + r^2 d\Omega^2
\end{aligned} \tag{5.220}$$

Defining new effective components:

$$\tilde{g} * aa(a, r) = g * aa + 2g_{ab}\frac{\partial b}{\partial a} + g_{bb} \left(\frac{\partial b}{\partial a} \right)^2 \tag{5.221}$$

$$\tilde{g} * ar(a, r) = g * ab\frac{\partial b}{\partial r} + g_{bb}\frac{\partial b}{\partial a}\frac{\partial b}{\partial r} \tag{5.222}$$

$$\tilde{g} * rr(a, r) = g * bb \left(\frac{\partial b}{\partial r} \right)^2 \tag{5.223}$$

$$ds^2 = g_{aa}(a, r)da^2 + g_{ar}(a, r)(dad r + drda) + g_{rr}(a, r)dr^2 + r^2 d\Omega^2 \tag{5.224}$$

Introducing time coordinate $t(a, r)$:

$$dt = \frac{\partial t}{\partial a}da + \frac{\partial t}{\partial r}dr \tag{5.225}$$

$$dt^2 = \left(\frac{\partial t}{\partial a} \right)^2 da^2 + 2 \left(\frac{\partial t}{\partial a} \right) \left(\frac{\partial t}{\partial r} \right) dad r + \left(\frac{\partial t}{\partial r} \right)^2 dr^2 \tag{5.226}$$

Targeting a diagonal metric:

$$ds^2 = m(t, r)dt^2 + n(t, r)dr^2 + r^2 d\Omega^2 \tag{5.227}$$

$$\begin{aligned}
m(t, r)dt^2 + n(t, r)dr^2 &= m(t, r) \left[\left(\frac{\partial t}{\partial a} \right)^2 da^2 + 2 \left(\frac{\partial t}{\partial a} \right) \left(\frac{\partial t}{\partial r} \right) dad r + \left(\frac{\partial t}{\partial r} \right)^2 dr^2 \right] + n(t, r)dr^2 \\
&= \left[m(t, r) \left(\frac{\partial t}{\partial a} \right)^2 \right] da^2 + 2 \left[m(t, r) \left(\frac{\partial t}{\partial a} \right) \left(\frac{\partial t}{\partial r} \right) \right] dad r \\
&+ \left[m(t, r) \left(\frac{\partial t}{\partial r} \right)^2 + n(t, r) \right] dr^2
\end{aligned} \tag{5.228}$$

Equating coefficients yields:

$$g_{aa} = m \left(\frac{\partial t}{\partial a} \right)^2 \tag{5.229}$$

$$g_{ar} = m \left(\frac{\partial t}{\partial a} \right) \left(\frac{\partial t}{\partial r} \right) \tag{5.230}$$

$$g_{rr} = m \left(\frac{\partial t}{\partial r} \right)^2 + n \tag{5.231}$$

This allows full diagonalization:

$$ds^2 = m(t, r)dt^2 + n(t, r)dr^2 + r^2d\Omega^2 \quad (5.232)$$

Defining positive functions:

$$m(t, r) = -e^{2\alpha(t, r)} \quad (5.233)$$

$$n(t, r) = e^{2\beta(t, r)} \quad (5.234)$$

$$ds^2 = -e^{2\alpha(t, r)}dt^2 + e^{2\beta(t, r)}dr^2 + r^2d\Omega^2 \quad (5.235)$$

By Ricci tensor conditions derived analogously to the static case:

$$R_{\mu\nu} = 0 \quad (5.236)$$

$$\frac{2}{r}\partial_t\beta(t, r) = 0 \quad (5.237)$$

$$\partial_t\beta(t, r) = 0 \quad (5.238)$$

$$\beta(t, r) = \beta(r) \quad (5.239)$$

$$R_{\theta\theta} = e^{-2\beta(r)} [r(\partial_r\beta(r) - \partial_r\alpha(t, r)) - 1] + 1 = 0 \quad (5.240)$$

$$-re^{-2\beta(r)}\partial_t\partial_r\alpha(t, r) = 0 \quad (5.241)$$

$$\alpha(t, r) = f(r) + g(t) \quad (5.242)$$

Absorbing the time dependence via a new coordinate $\tilde{t}$:

$$d\tilde{t}^2 = e^{2g(t)}dt^2 \quad (5.243)$$

$$\tilde{t}(t) = \int e^{g(t)}dt \quad (5.244)$$

$$ds^2 = -e^{2f(r)}d\tilde{t}^2 + e^{2\beta(r)}dr^2 + r^2d\Omega^2 \quad (5.245)$$

Dropping the tilde, we recover the static form, proving Birkhoff's theorem:

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2d\Omega^2 \quad (5.246)$$

5.3 Coordinate vs. Curvature Singularities

The Kretschmann scalar $K = R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}$ indicates true singularities:

$$K = R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}. \quad (5.247)$$

In an orthonormal tetrad basis:

$$K = \sum_{\mu, \nu, \rho, \sigma} R^{\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}}R_{\hat{\mu}\hat{\nu}\hat{\rho}\hat{\sigma}}. \quad (5.248)$$

$$\begin{aligned} K &= 4 \times (R_{\hat{t}\hat{r}\hat{t}\hat{r}})^2 \\ &+ 4 \times (R_{\hat{t}\hat{\theta}\hat{t}\hat{\theta}})^2 + 4 \times (R_{\hat{t}\hat{\phi}\hat{t}\hat{\phi}})^2 \\ &+ 4 \times (R_{\hat{r}\hat{\theta}\hat{r}\hat{\theta}})^2 + 4 \times (R_{\hat{r}\hat{\phi}\hat{r}\hat{\phi}})^2 \\ &+ 4 \times (R_{\hat{\theta}\hat{\phi}\hat{\theta}\hat{\phi}})^2. \end{aligned} \quad (5.249)$$

$$\begin{aligned}
K &= 4 \left(-\frac{2GM}{r^3} \right)^2 + 4 \left(\frac{GM}{r^3} \right)^2 + 4 \left(\frac{GM}{r^3} \right)^2 + 4 \left(-\frac{GM}{r^3} \right)^2 + 4 \left(-\frac{GM}{r^3} \right)^2 + 4 \left(\frac{2GM}{r^3} \right)^2 \\
&= 4 \left(\frac{4G^2M^2}{r^6} \right) + 4 \left(\frac{G^2M^2}{r^6} \right) + 4 \left(\frac{G^2M^2}{r^6} \right) + 4 \left(\frac{G^2M^2}{r^6} \right) + 4 \left(\frac{G^2M^2}{r^6} \right) + 4 \left(\frac{4G^2M^2}{r^6} \right) \\
&= \frac{G^2M^2}{r^6} [16 + 4 + 4 + 4 + 4 + 16] \\
K &= \frac{48G^2M^2}{r^6}.
\end{aligned} \tag{5.250}$$

$K \rightarrow \infty$ at $r = 0$ indicates a true curvature singularity. At $r = 2GM$, K is finite:

$$K(r = 2GM) = \frac{48G^2M^2}{(2GM)^6} = \frac{48G^2M^2}{64G^6M^6} = \frac{3}{4G^4M^4}. \tag{5.251}$$

This confirms $r = 2GM$ is merely a coordinate singularity.

5.4 Geodesics of the Schwarzschild Geometry

$$ds^2 = - \left(1 - \frac{2GM}{r} \right) dt^2 + \left(1 - \frac{2GM}{r} \right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2). \tag{5.252}$$

$$\mathcal{L} = \frac{1}{2} g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}. \tag{5.253}$$

$$\mathcal{L} = \frac{1}{2} \left[- \left(1 - \frac{2GM}{r} \right) \dot{t}^2 + \left(1 - \frac{2GM}{r} \right)^{-1} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2 \right], \tag{5.254}$$

Constants of motion E and L :

$$-E \equiv p_t = \frac{\partial \mathcal{L}}{\partial \dot{t}} = - \left(1 - \frac{2GM}{r} \right) \dot{t} \implies E = \left(1 - \frac{2GM}{r} \right) \frac{dt}{d\lambda}. \tag{5.255}$$

$$L \equiv p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = r^2 \sin^2 \theta \frac{d\phi}{d\lambda}. \tag{5.256}$$

Restricting to the equatorial plane $\theta = \pi/2$:

$$\theta = \frac{\pi}{2}, \quad \sin \theta = 1, \quad \dot{\theta} = 0. \tag{5.257}$$

$$L = r^2 \frac{d\phi}{d\lambda}. \tag{5.258}$$

Using the normalization $g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = -\epsilon$:

$$- \left(1 - \frac{2GM}{r} \right) \left(\frac{dt}{d\lambda} \right)^2 + \left(1 - \frac{2GM}{r} \right)^{-1} \left(\frac{dr}{d\lambda} \right)^2 + r^2 \left(\frac{d\phi}{d\lambda} \right)^2 = -\epsilon. \tag{5.259}$$

$$\frac{dt}{d\lambda} = E \left(1 - \frac{2GM}{r} \right)^{-1}, \tag{5.260}$$

$$\frac{d\phi}{d\lambda} = \frac{L}{r^2}. \tag{5.261}$$

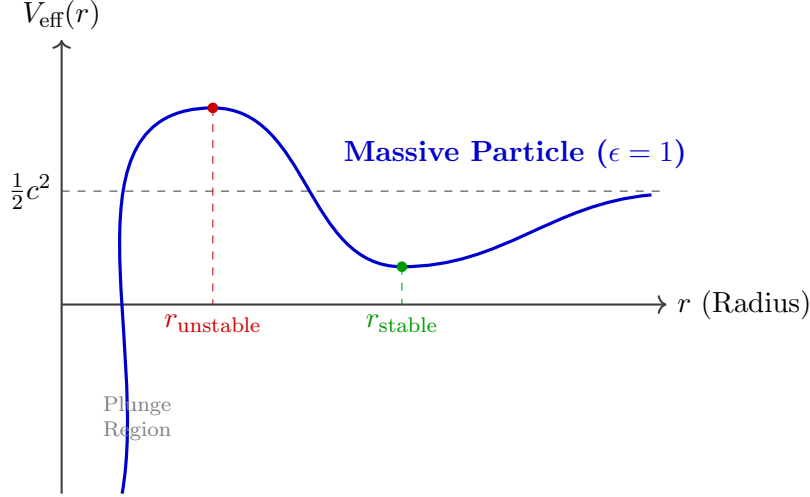
$$- \left(1 - \frac{2GM}{r} \right) \left[E \left(1 - \frac{2GM}{r} \right)^{-1} \right]^2 + \left(1 - \frac{2GM}{r} \right)^{-1} \left(\frac{dr}{d\lambda} \right)^2 + r^2 \left(\frac{L}{r^2} \right)^2 = -\epsilon. \tag{5.262}$$

$$-E^2 \left(1 - \frac{2GM}{r} \right)^{-1} + \left(1 - \frac{2GM}{r} \right)^{-1} \left(\frac{dr}{d\lambda} \right)^2 + \frac{L^2}{r^2} = -\epsilon. \tag{5.263}$$

$$-E^2 + \left(\frac{dr}{d\lambda}\right)^2 + \left(1 - \frac{2GM}{r}\right) \frac{L^2}{r^2} = -\epsilon \left(1 - \frac{2GM}{r}\right). \quad (5.264)$$

$$\frac{1}{2} \left(\frac{dr}{d\lambda}\right)^2 + V_{\text{eff}}(r) = \mathcal{E}, \quad (5.265)$$

$$V_{\text{eff}}(r) = \frac{1}{2}\epsilon - \epsilon \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \quad (5.266)$$



For photons ($\epsilon = 0$):

$$V_{\text{eff}}(r) = \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \quad (5.267)$$

$$\frac{dV_{\text{eff}}}{dr} = -\frac{L^2}{r^3} + \frac{3GML^2}{r^4}. \quad (5.268)$$

$$\frac{L^2}{r^3} = \frac{3GML^2}{r^4} \implies r_c = 3GM. \quad (5.269)$$

$$\frac{d^2V_{\text{eff}}}{dr^2} = \frac{3L^2}{r^4} - \frac{12GML^2}{r^5}. \quad (5.270)$$

$$\left. \frac{d^2V_{\text{eff}}}{dr^2} \right|_{*r=3GM} = -\frac{L^2}{81G^4M^4}. \quad (5.271)$$

For massive particles ($\epsilon = 1$):

$$V_{\text{eff}}(r) = \frac{1}{2} - \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3}. \quad (5.272)$$

$$\frac{dV_{\text{eff}}}{dr} = \frac{GM}{r^2} - \frac{L^2}{r^3} + \frac{3GML^2}{r^4} = 0. \quad (5.273)$$

$$r^2 - \left(\frac{L^2}{GM}\right)r + 3L^2 = 0. \quad (5.274)$$

$$r_c = \frac{L^2 \pm \sqrt{L^4 - 12G^2M^2L^2}}{2GM}. \quad (5.275)$$

Critical threshold angular momentum for orbits:

$$L_c = \sqrt{12}GM. \quad (5.276)$$

Innermost Stable Circular Orbit (ISCO):

$$r_{\text{ISCO}} = \frac{12G^2M^2}{2GM} = 6GM. \quad (5.277)$$

5.5 Experimental Tests of General Relativity

5.5.1 The Precession of Perihelia

$$\frac{1}{2} \left(\frac{dr}{d\lambda} \right)^2 + \frac{1}{2} - \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3} = \frac{1}{2} E^2. \quad (5.278)$$

$$\frac{dr}{d\lambda} = \frac{dr}{d\phi} \frac{d\phi}{d\lambda} = \frac{L}{r^2} \frac{dr}{d\phi}. \quad (5.279)$$

$$\frac{L^2}{r^4} \left(\frac{dr}{d\phi} \right)^2 + 1 - \frac{2GM}{r} + \frac{L^2}{r^2} - \frac{2GML^2}{r^3} = E^2. \quad (5.280)$$

Substituting $u = 1/r$:

$$\frac{dr}{d\phi} = -r^2 \frac{du}{d\phi}. \quad (5.281)$$

$$L^2 \left(\frac{du}{d\phi} \right)^2 + 1 - 2GMu + L^2 u^2 - 2GML^2 u^3 = E^2. \quad (5.282)$$

$$2L^2 \left(\frac{du}{d\phi} \right) \frac{d^2 u}{d\phi^2} - 2GM \frac{du}{d\phi} + 2L^2 u \frac{du}{d\phi} - 6GML^2 u^2 \frac{du}{d\phi} = 0. \quad (5.283)$$

$$\frac{d^2 u}{d\phi^2} + u = \frac{GM}{L^2} + 3GMu^2. \quad (5.284)$$

Scaling $x = L^2 u / GM$:

$$x = \frac{L^2}{GM} u = \frac{L^2}{GM r}. \quad (5.285)$$

$$\frac{d^2 x}{d\phi^2} - 1 + x = \frac{3G^2 M^2}{L^2} x^2. \quad (5.286)$$

Using perturbative expansion $x = x_0 + x_1$:

$$x(\phi) = x_0(\phi) + x_1(\phi). \quad (5.287)$$

$$\text{Zeroth order: } \frac{d^2 x_0}{d\phi^2} + x_0 = 1, \quad (5.288)$$

$$\text{First order: } \frac{d^2 x_1}{d\phi^2} + x_1 = \alpha x_0^2. \quad (5.289)$$

$$x_0(\phi) = 1 + e \cos \phi, \quad (5.290)$$

$$\frac{d^2 x_1}{d\phi^2} + x_1 = \alpha(1 + e \cos \phi)^2 = \alpha(1 + 2e \cos \phi + e^2 \cos^2 \phi). \quad (5.291)$$

$$\frac{d^2 x_1}{d\phi^2} + x_1 = \alpha \left[\left(1 + \frac{1}{2} e^2 \right) + 2e \cos \phi + \frac{1}{2} e^2 \cos 2\phi \right]. \quad (5.292)$$

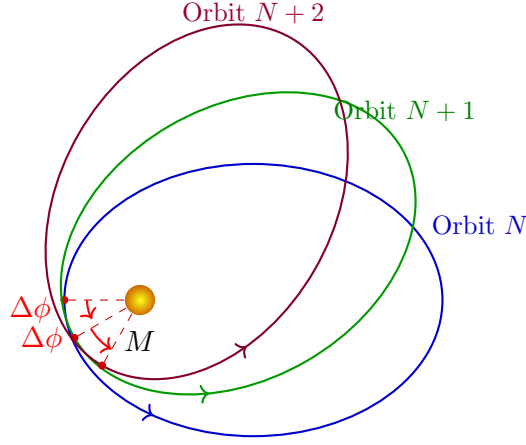
$$\frac{d}{d\phi} (D\phi \sin \phi) = D \sin \phi + D\phi \cos \phi, \quad (5.293)$$

$$\frac{d^2}{d\phi^2} (D\phi \sin \phi) = 2D \cos \phi - D\phi \sin \phi. \quad (5.294)$$

$$x_1(\phi) = \alpha \left[\left(1 + \frac{1}{2} e^2 \right) + e\phi \sin \phi - \frac{1}{6} e^2 \cos 2\phi \right]. \quad (5.295)$$

$$x(\phi) \approx 1 + e \cos \phi + \alpha e\phi \sin \phi. \quad (5.296)$$

$$x(\phi) \approx 1 + e \cos(\phi - \alpha\phi) = 1 + e \cos[(1 - \alpha)\phi]. \quad (5.297)$$



$$(1 - \alpha)\phi_p = 2\pi \quad \Rightarrow \quad \phi_p = \frac{2\pi}{1 - \alpha}. \quad (5.298)$$

$$\phi_p \approx 2\pi(1 + \alpha) = 2\pi + 2\pi\alpha. \quad (5.299)$$

$$\Delta\phi = 2\pi\alpha = \frac{6\pi G^2 M^2}{L^2}. \quad (5.300)$$

$$\Delta\phi = \frac{6\pi GM}{c^2 a(1 - e^2)} \text{ radians per orbit.} \quad (5.301)$$

5.5.2 Gravitational Redshift

$$g_{tt}(U^t)^2 = -1 \quad \Rightarrow \quad -\left(1 - \frac{2GM}{r}\right)(U^t)^2 = -1. \quad (5.302)$$

$$U^\mu = \left(\left(1 - \frac{2GM}{r}\right)^{-1/2}, 0, 0, 0 \right). \quad (5.303)$$

$$\omega = -g_{\mu\nu}p^\mu U^\nu. \quad (5.304)$$

$$\omega = -g_{tt}p^t U^t = -\left[-\left(1 - \frac{2GM}{r}\right) \right] \left(\frac{dt}{d\lambda} \right) \left(1 - \frac{2GM}{r}\right)^{-1/2}. \quad (5.305)$$

$$E = \left(1 - \frac{2GM}{r}\right) \frac{dt}{d\lambda} \quad \Rightarrow \quad \frac{dt}{d\lambda} = E \left(1 - \frac{2GM}{r}\right)^{-1}. \quad (5.306)$$

$$\omega = \left(1 - \frac{2GM}{r}\right) \left[E \left(1 - \frac{2GM}{r}\right)^{-1} \right] \left(1 - \frac{2GM}{r}\right)^{-1/2}$$

$$\omega = E \left(1 - \frac{2GM}{r}\right)^{-1/2}. \quad (5.307)$$

$$\frac{\omega_2}{\omega_1} = \frac{E \left(1 - \frac{2GM}{r_2}\right)^{-1/2}}{E \left(1 - \frac{2GM}{r_1}\right)^{-1/2}} = \left(\frac{1 - \frac{2GM}{r_1}}{1 - \frac{2GM}{r_2}} \right)^{1/2}. \quad (5.308)$$

Weak field limit:

$$\begin{aligned} \frac{\omega_2}{\omega_1} &= \left(1 - \frac{2GM}{r_1}\right)^{1/2} \left(1 - \frac{2GM}{r_2}\right)^{-1/2} \\ &\approx \left(1 - \frac{GM}{r_1}\right) \left(1 + \frac{GM}{r_2}\right) \\ &\approx 1 - \frac{GM}{r_1} + \frac{GM}{r_2} + \mathcal{O}\left(\frac{G^2 M^2}{r^2}\right). \end{aligned} \quad (5.309)$$

$$\frac{\omega_2}{\omega_1} \approx 1 + \Phi(r_1) - \Phi(r_2) = 1 - \Delta\Phi. \quad (5.310)$$

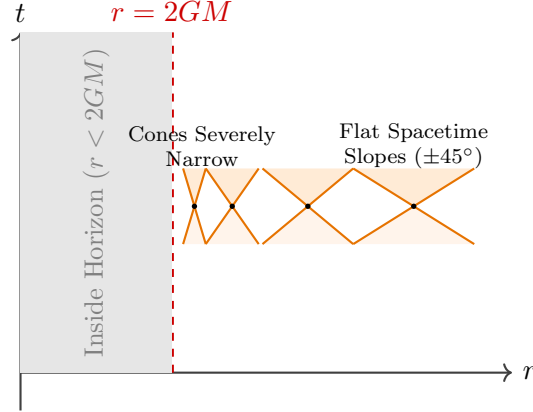
5.6 Schwarzschild Black Holes and Causal Structure

5.6.1 Causal Structure in Schwarzschild Coordinates

$$0 = - \left(1 - \frac{2GM}{r} \right) dt^2 + \left(1 - \frac{2GM}{r} \right)^{-1} dr^2. \quad (5.311)$$

$$\frac{dt}{dr} = \pm \left(1 - \frac{2GM}{r} \right)^{-1}, \quad (5.312)$$

$$\lim_{r \rightarrow 2GM} \frac{dt}{dr} = \pm \infty. \quad (5.313)$$



5.6.2 The Tortoise Coordinate

$$dt = \pm \frac{dr}{1 - 2GM/r} \implies \int dt = \pm \int \frac{r}{r - 2GM} dr. \quad (5.314)$$

$$\pm t = r + 2GM \ln \left| \frac{r}{2GM} - 1 \right| + \text{constant}. \quad (5.315)$$

$$r^* \equiv r + 2GM \ln \left| \frac{r}{2GM} - 1 \right|. \quad (5.316)$$

$$dr^* = \left(1 + \frac{2GM}{r - 2GM} \right) dr = \left(\frac{r}{r - 2GM} \right) dr = \left(1 - \frac{2GM}{r} \right)^{-1} dr. \quad (5.317)$$

$$\left(1 - \frac{2GM}{r} \right)^{-1} dr^2 = \left(1 - \frac{2GM}{r} \right) (dr^*)^2. \quad (5.318)$$

$$ds^2 = \left(1 - \frac{2GM}{r} \right) [-dt^2 + (dr^*)^2] + r^2 d\Omega^2, \quad (5.319)$$

5.6.3 Eddington-Finkelstein Coordinates

$$v \equiv t + r^*. \quad (5.320)$$

$$dt = dv - dr^* = dv - \left(1 - \frac{2GM}{r} \right)^{-1} dr. \quad (5.321)$$

$$\begin{aligned} ds^2 &= - \left(1 - \frac{2GM}{r} \right) \left[dv - \left(1 - \frac{2GM}{r} \right)^{-1} dr \right]^2 + \left(1 - \frac{2GM}{r} \right)^{-1} dr^2 + r^2 d\Omega^2 \\ &= - \left(1 - \frac{2GM}{r} \right) \left[dv^2 - 2 \left(1 - \frac{2GM}{r} \right)^{-1} dv dr + \left(1 - \frac{2GM}{r} \right)^{-2} dr^2 \right] \\ &\quad + \left(1 - \frac{2GM}{r} \right)^{-1} dr^2 + r^2 d\Omega^2. \end{aligned} \quad (5.322)$$

$$ds^2 = - \left(1 - \frac{2GM}{r}\right) dv^2 + 2dvdr - \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (5.323)$$

$$ds^2 = - \left(1 - \frac{2GM}{r}\right) dv^2 + 2dvdr + r^2 d\Omega^2. \quad (5.324)$$

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2GM}{r}\right) & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix} \quad (5.325)$$

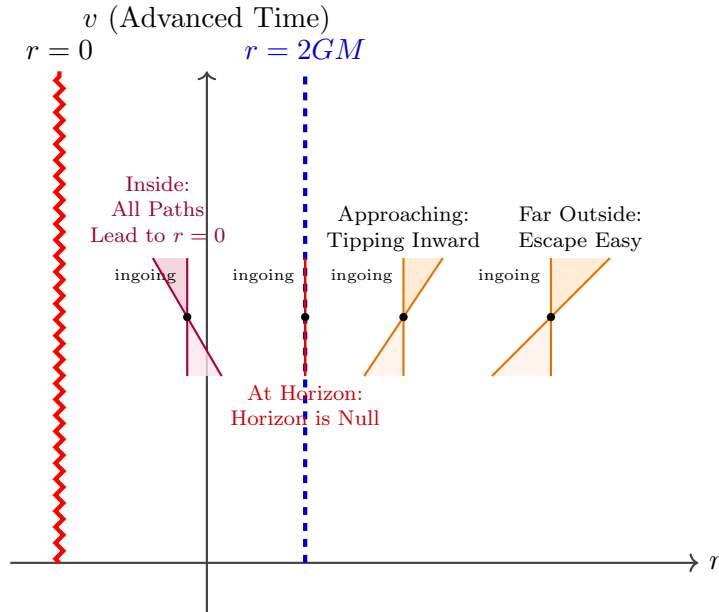
$$g = \det(g_{\mu\nu}) = -(0 - (1)(1))(r^2)(r^2 \sin^2 \theta) = -r^4 \sin^2 \theta. \quad (5.326)$$

5.6.4 The Event Horizon and the Tipping of Light Cones

$$0 = - \left(1 - \frac{2GM}{r}\right) dv^2 + 2dvdr = dv \left[- \left(1 - \frac{2GM}{r}\right) dv + 2dr \right]. \quad (5.327)$$

$$\text{Ingoing null geodesics: } \frac{dv}{dr} = 0 \implies v = \text{constant}, \quad (5.328)$$

$$\text{Outgoing null geodesics: } \frac{dv}{dr} = 2 \left(1 - \frac{2GM}{r}\right)^{-1}. \quad (5.329)$$



The surface $r = 2GM$ forms an event horizon because all future light cones are tilted inward, making escape impossible.

$$\frac{1}{2}mv_{\text{esc}}^2 = \frac{GMm}{r} \implies v_{\text{esc}} = \sqrt{\frac{2GM}{r}}. \quad (5.330)$$